

Need for Speed, but How Much Does It Cost?
Unpacking the Fee-Speed Relationship in Bitcoin Transactions

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Abstract: The growing popularity of blockchain-based cryptocurrencies is driven by the flexibility in transaction fee offerings, among other factors. To achieve service-level differentiation among their users, many cryptocurrencies allow users to “name your own price”, giving rise to a large variation in fee offerings and hence, variation in confirmation times. Yet, the time it takes a cryptocurrency transaction to be confirmed in the blockchain is not only affected by the fee offered, but also by the contemporaneous congestion level and the inherent randomness in the verification process. Although it is generally expected that higher fees lead to quicker confirmation, the uniqueness of the cryptocurrency setting adds important nuances to the fee-speed relationship. Using Bitcoin – the original and most heavily used cryptocurrency by far – as our empirical context, we stylize the transaction confirmation processes, propose a theoretical framework that maps the causal path from fee to speed, and estimate this framework using Bitcoin transaction data under periods of high volatility. Our results show strong evidence for two characteristics of fee’s impact: congestion dependence and tail shrinkage. Our finding that the speed acceleration effect of fee is particularly strong on the tail of the confirmation time distribution motivates a target service level approach to fee recommendation. To put this finding into practice, we develop an efficient computational procedure that helps Bitcoin users accurately estimate fees based on their confirmation delay preferences. We discuss the implications of our analyses on future cryptocurrency development and the long-term adoption of this revolutionary technology.

Key words: *congested service system; service speed; service fee; priority queues; name your own price (NYOP); pay what you want (PWYW); blockchain; cryptocurrency*

1. Introduction and Literature

Blockchain is an emerging data management technology used for developing and maintaining distributed ledgers (Babich and Hilary, 2020). Considered an Industry 4.0 era disruptive technology (Choi et al., 2022; Olsen and Tomlin, 2020), it proliferates in various areas ranging from logistics and supply chain management (Chod et al., 2020; Choi and Luo, 2019; Hastig and Sodhi, 2020) to crowdsourcing (Gan et al., 2021; Tsoukalas and Falk, 2020). Yet, the financial services sector – particularly the cryptocurrency field, remains the most established and widely used domain of this technology (Zhao et al., 2022). A cryptocurrency is a digital currency system based on peer-to-peer network protocols and cryptographic methods that is used to facilitate the transfer of financial ownership. In 2020, the average daily market capitalization of cryptocurrencies has been over 300 billion USD¹. Given the sheer size of this market and the increasing popularity of cryptocurrencies, it is unsurprising to observe a keen academic interest in studying this technology. In particular, recent cryptocurrency studies have concentrated on price prediction (Bouri et al., 2019b; Li and Wang, 2017), price manipulation (Fry, 2018; Gandal et al., 2018), market participant behavior (Bouri et al., 2019a; Roşu and Saleh, 2021), and market design (Huberman et al., 2019).

However, so far, less attention has been given to studying operational fundamentals such as the transaction fee mechanism that regulates the confirmation times in most cryptocurrencies (e.g., Bitcoin). From a modeling standpoint, the confirmation process in a cryptocurrency blockchain can be viewed as a priority queueing system with priorities determined by transaction fees (Kawase and Kasahara, 2020). Fees are payments made by the users of a cryptocurrency in order to facilitate transaction processing (Lu et al., 2021). Users are permitted to offer any fee amount (i.e., flexible fees) to compete for the computing resources provided by miners – a special type of users who provide the computing power to validate a transaction. All else equal, offering higher fees is expected to increase the speed that a transaction will be processed because economically rational miners choose to validate transactions with higher fees first.

¹ <https://coinmarketcap.com>

With the broader adoption of cryptocurrency technology and the increasing volume of transactions arriving to cryptocurrency networks, the amount of fees users pay continue to grow. In 2020, the average fee in the Bitcoin system² has been 2.87 USD per transaction and the total amount of transaction fees paid to miners was over 326 million USD³. Besides its increasing magnitude, the volatility of transaction fees has also been dramatic. For instance, as of January 2021, the average fee per transaction has jumped to 11.35 USD. This spike was fueled by the surge in transaction volume occurring on Bitcoin exchange sites leading to congestion in the network (Deka, 2021). It is important to note that such an increase in fees did not necessarily translate to faster processing times as during the same period, the average time for a transaction to be confirmed in the system also increased by 58% – from 74.6 minutes to 117.9 minutes⁴. This seemingly counter-intuitive outcome is caused by a design feature that limits the maximum throughput on the Bitcoin network to 7 transactions per second, regardless of the amount of fees offered (Croman et al., 2016). (See Section 2.2 for details about the throughput limit).

The aforementioned statistics allude to two facts. First, the average fee paid per transaction is correlated with the adoption of this technology and will continue to increase as long as the throughput limit embedded in system protocols stays in effect. Second, there exists an inherent uncertainty in the appraisal of fees as the same amount of fee might lead to vastly different service outcomes (in terms of transaction confirmation speed) at two different times. Basu et al. (2019) argue that the day-to-day volatility in transaction fees makes it very difficult to determine the proper fee amount in accordance with the confirmation time desired by users and hence leads to poor service quality. The goal of this paper is to address this concern by asking: *what is the relationship between transaction fee and transaction confirmation speed in the Bitcoin cryptocurrency system?* In particular, we focus on the fee's nuanced impact under different service levels

² Bitcoin is the original and most popular cryptocurrency system in use today with over 60% of the cryptocurrency market share in 2020.

³ <https://www.blockchain.com/charts>

⁴ <https://www.blockchain.com/charts/avg-confirmation-time>

and varying degrees of network congestion. To the best of our knowledge, there is no existing empirical research that systematically investigated the impact of fee on speed at the transaction-level⁵.

Following the Blockchain-related research directions outlined in Babich and Hilary (2020), we argue that the Operations Management (OM) field has a lot to contribute to this particular research question. Specifically, the implementation of the blockchain as a priority queueing system allows for service-level differentiation based on the transaction fees offered by the users (i.e., the price of a transaction). When a service system is congested and customers have heterogeneous waiting costs, “*offering multiple service grades that each render a different delay distribution at a different price*” improves both perceived customer satisfaction and service provider profit (Van Mieghem 2000 p.1249). To establish different priority levels among the customers, many traditional services rely on a centralized authority (e.g., seller or owner) that sets the prices and/or the speed of the service delivery. On the other hand, Bitcoin’s transaction confirmation service, while also highly susceptible to system congestion, is distinctively decentralized – i.e., there is no central entity (e.g., platform operator) to control pricing or service speed. To better highlight this distinction, we categorize an array of common services using two dimensions: price setting party and service speed uncertainty, as shown in Table 1.

Table 1. Price-Setting and Delivery Speed Uncertainty Characteristics of Common Services

	Speed of Service Delivery	
	Low Uncertainty	High Uncertainty
Seller Sets Price	Bank wire transfer, Amazon Prime, parcel delivery (UPS and FedEx)	CLEAR airport security check point, Disney FASTPASS
Buyer Sets Price	Surge pricing (DiDi and Uber)	Bitcoin transaction fee

In most service settings, sellers set the service price and the service speed uncertainty is very low. For instance, Amazon Prime is a membership service that enables its subscribers to receive faster delivery of their purchases in return for an annual fee. Amazon sets the membership price and users normally are

⁵ In our literature survey, the only similar study that we identified is by Easley et al. (2019), who studied the impact of speed on fee using the aggregate daily-level data.

guaranteed to receive their products in two days. The fee-schedule relations at logistics service providers such as UPS and FedEx (e.g., 2nd day vs economy shipping) are other examples of service with sellers setting the price and low speed uncertainty. Accordingly, we group such services under the “Low Uncertainty” & “Seller Sets the Price” quadrant in Table 1. There are also some settings where sellers set the price without service speed guarantees. For example, CLEAR is a program that provides expedited identity verification service by using priority lines at participating airports. Enrollees of the CLEAR program pay an annual fee that is set by the seller but their waiting times at the airport are highly uncertain and depend on how many people are using the service simultaneously. Therefore, we categorize this service under the “Speed Uncertainty High” & “Seller Sets Price” quadrant. Next, we consider the surge pricing functionality commonly employed by ride hailing services such as Uber and DiDi. During peak demand and service congestion, surge pricing allows customers to offer additional fees, similar to flexible Bitcoin transaction fees, to receive faster service (e.g., booking the next available ride). However, different from Bitcoin, bids are accepted/rejected almost instantaneously in surge pricing and the system can estimate the arrival time of the ride relatively accurately – hence such services demonstrate low speed uncertainty. Finally, we move on to the 4th and the last quadrant, in which our context (Bitcoin) occupies a unique space. The Bitcoin transaction service is different from other types of services because it requires users to set prices for their own transactions all the time (not just during congested times), which leads to large variation in fee offerings. As mentioned earlier, this characteristic is not an unforeseen deficiency, but rather an inherent design feature that allows low willingness-to-wait users the flexibility to offer extra transaction fees with the expectation of moving up in the service queue. However, there are no speed guarantees as the time it takes for a transaction to be confirmed by miners is not only affected by the focal user’s fee amount, but also the fee offerings of other adjacent past and future users in the system and the inherent randomness in the verification process.

From a practical standpoint, our study aims to bring transparency into the operations of Bitcoin transaction confirmation process. This high-level goal is in line with the recent trend in recognizing the value that operational transparency brings to both customers and service providers (Buell et al., 2017; Buell

and Norton, 2011). More specifically, despite the general expectation of a positive fee-speed relationship (Böhme et al., 2015), the magnitude of this relationship is sensitive to system dynamics and not transparent to users, hindering their ability to accurately gauge the true value of transaction fees. Analogous to delay announcements in service queues (Akşın et al., 2017; Yu et al., 2016), the information on confirmation time distribution (given a fee amount) should improve perceived service quality. Such transparency is also expected to benefit miners, in addition to Bitcoin users. This is because miners' minting revenue will gradually decrease to zero in the long run as dictated by the Bitcoin protocol, which in turn places more reliance on the transaction fee revenue to incentivize mining activities.

Despite Bitcoin's first introduction by Nakamoto (2008) over a decade ago and hyped media coverage since then (Philippas et al., 2019), fundamental questions such as how the Bitcoin transaction confirmation process works and what the mechanisms underlying fee's impact on speed are remain unanswered. The institutional details presented in this paper represent a crucial first step in unpacking the underpinnings of the transaction fee mechanism using an OM perspective. To delineate the fee-speed relationship in detail, we develop a theoretical framework that traces the journey of a transaction from being submitted to the service queue (i.e., pending transactions referred to as the memory pool) to being included in a future block. Our framework shows that confirmation time is jointly determined by the inter-arrival time between blocks and the probability of being confirmed in future blocks, and the causal effect of fee on speed is transmitted via the latter. Specifically, by considering all pending and projected transactions that are potentially competing for a spot in a future block, we reveal that the impact of fee on speed propagates through two priority pools (visible and invisible) in the queue. In other words, it is not purely the nominal fee but also the priority ranking of a transaction in the queue that determines the confirmation times. Furthermore, we find that increasing fees have a much larger impact on the tail of the confirmation time distribution than on the mean – a phenomenon that we refer to as the tail shrinkage effect. This is insightful for Bitcoin users who care more about a certain service level target than the expected confirmation time. To demonstrate how our results can be used to improve fee recommendation, we develop a computationally efficient three-step procedure that calculates the distribution of confirmation time under a wide range of fee inputs. For a

user with a desired transaction service level such as “*confirmed in the next 60 minutes with a 90% likelihood*”, our computational procedure recommends a best fee to achieve this goal.

2. Institutional Background

In this section, we first review the relevant OM pricing literature while highlighting the unique features of the transaction pricing (i.e., transaction fee setting) mechanism in Bitcoin. Then, we provide an overview of the bitcoin mining process and discuss the details of the transaction confirmation procedure on the Bitcoin network.

2.1. Relevant OM Pricing Literature

From a theoretical standpoint, our study brings together two longstanding OM literature streams – Name-Your-Own-Price (NYOP) and congested service pricing – in a novel context. Since pioneered by Priceline.com in 1998, NYOP has been widely adopted in the hospitality industry (e.g., airlines and hotels) as a strategy to clear excess capacity (Anderson, 2009). Its popularity has later propagated into a variety of other sectors such as online consumer electronics retailing (Hann and Terwiesch, 2003; Terwiesch et al., 2005), restaurant meals, and designer fashion accessories (Zeithammer, 2015). Despite the nuances in implementation (e.g., how many times a customer can name the price), Fay and Zeithammer (2016) succinctly summarize how NYOP works as “if a bid exceeds the concealed threshold price [set by the seller], the consumer receives the product at her bid price.” A more lenient variant, termed Pay-What-You-Want (PWYW), eliminates this threshold (Gneezy et al., 2010). That is, a customer can still obtain the product/service even if she chooses not to pay. PWYW examples include amusement park souvenirs (Gneezy et al., 2010), crowd-sourced information repositories such as Wikipedia (Mak et al., 2015), museum admissions (Schons et al., 2014), and music albums (El Harbi et al., 2014). The flexibility Bitcoin users enjoy in choosing transaction fees resembles NYOP and PWYW to a large degree. Similar to PWYW, users will have their transactions confirmed in the blockchain eventually even if they “name a zero price”. However, unlike NYOP and PWYW, where different prices generally lead to the same product/service (e.g., visitors who pay \$5 and \$50 for a museum admission see the same collections), Bitcoin users with higher transaction fees are likely to receive better service quality, measured by confirmation speed.

The fundamental reason behind this contrast is that Bitcoin's service delivery is highly contingent on system congestion; and transaction fees grant users priority on obtaining the service, which naturally brings us to the second related OM literature stream – pricing congested services. Early studies in this line of research focus on public infrastructure such as highway tolls (Naor, 1969) and electric power supply (Viswanathan and Edison, 1989). Cachon and Feldman (2011) compare the pay-per-use menu pricing with a subscription pricing and show situations where the latter is more profitable. With the emergence of on-demand service platforms such as Uber (taxi hailing), DoorDash (food delivery), and Postmates (general consumer goods delivery), managing surge pricing (Cachon et al., 2017) and coordinating pricing with labor wage (Taylor, 2018) have emerged as promising research avenues. A common feature carried throughout the evolution of this long literature is that it is always the supplier – centralized service provider, platform operator, or agents working on platforms – setting the price (see Hassin and Haviv (2003) for a survey of early studies and Table 1 in Taylor (2018) for platform examples). On the other hand, as mentioned earlier, Bitcoin's transaction confirmation service is highly decentralized and there is no central authority to control pricing and/or speed. In summary, the ingenious design of the Bitcoin protocol brings together two distinct streams: it matches the predominantly seller-controlled congested service pricing problem with a customer-participative pricing mechanism often used in the hospitality industry to clear excess capacity.

2.2. Bitcoin Mining

Bitcoin mining refers to the process of validating and compiling transactions into collections called blocks, which are then added to the blockchain by linking them with the previously added blocks (Nakamoto, 2008). These operations are performed by a special type of users called miners, who use their computational power to generate and publish blocks on the blockchain. Briefly speaking, chaining a new block to the existing blockchain consists of three steps: assembling a number of unconfirmed transactions into a block template (assembly), solving the proof-of-work problem through constant trial-and-error (hashing), and propagating the “mined block” across the internet (propagation). In practice, the assembly step consumes trivial amount of time, since the transaction-picking heuristic is pre-coded. Hashing makes

up the vast majority of the time consumption and propagation is responsible for the minor rest – they together define block discovery period – i.e., Inter-Block Time (*IBT*). It is also important to note that only the first miner who successfully solves the proof-of-work problem has the right to publish his/her block (hence collect rewards) and the proof-of-work problem resets to a new random solution every time a new block is added to the blockchain. The difficulty of the proof-of-work problem is periodically adjusted by the Bitcoin protocol so that the long-term average of *IBT* stays as 10 minutes per block, regardless of the collective computational power added to the network.

The heavy computational requirements for hashing and hence the resultant operating costs associated with the mining activity necessitate the existence of a financial reward scheme to motivate the miners to participate in the network (Basu et al., 2019). In Bitcoin and its variants (e.g., Bitcoin Cash and Litecoin), two such incentive mechanisms are the minting revenue and the transaction fee revenue.

First, each miner that successfully discovers a block and chains it to the blockchain is awarded a number of newly minted bitcoins. This is called the minting revenue, which is the only way of inserting new bitcoins into circulation (Lavi et al., 2019). Under the Bitcoin protocol, the number of bitcoins awarded per new block originated at 50 and decreases by half every 210,000 blocks – approximately 4 years (Li and Wang, 2017). As of October 2021, minting award is 6.25 bitcoins/block. The same rate is expected to carry through until 2024. This protocol rule also means that the total number of bitcoins in circulation will be capped at approximately 21 million. Figure 1 illustrates the historical and the projected growth of the total number of bitcoins in circulation.

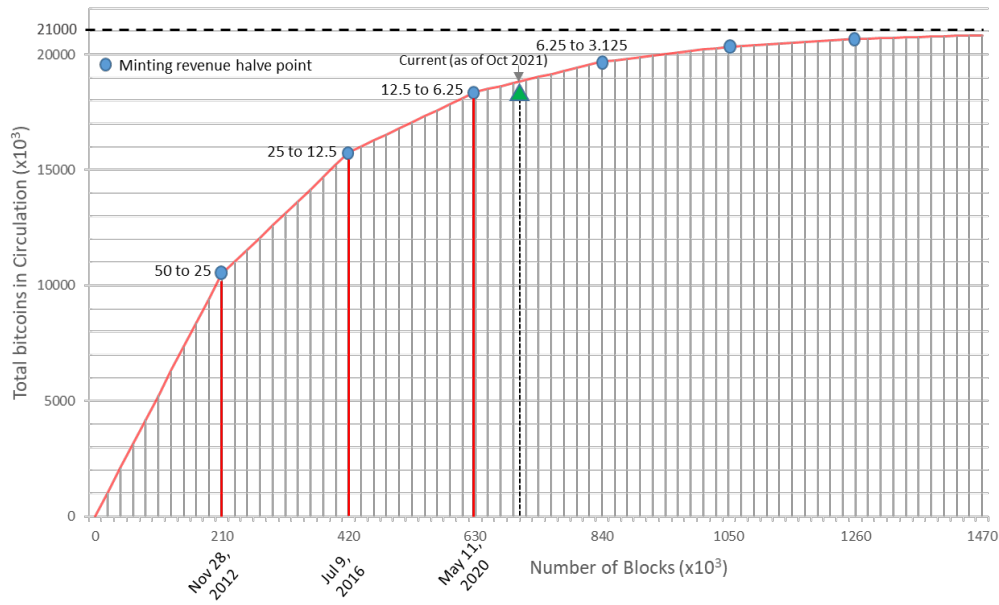


Figure 1. Evolution of Minting Revenue and Total Bitcoin Circulation

Second, to supplement decreasing minting rewards, the Bitcoin protocol also uses transaction fee revenue as an additional reward mechanism. In contrast with other popular gig economy platforms, where the platform operator takes a commission from the payments customers make to service providers (Cachon et al., 2017), Bitcoin runs automatically per protocol design and hence miners collect the entire sum of transaction fees in a block offered by senders of these transactions. During 2020, the total transaction fees paid to miners per day fluctuated between \$71,819 and \$4,605,834⁶.

The long-term effect of transitioning from minting-dominant to fee-dominant miner revenue structure has been the subject of a number of recent analytical studies. Carlsten et al. (2016) model an environment where the transaction fee is the sole reward mechanism and find that the Bitcoin system is considerably more vulnerable without the help of a fixed block reward regime. Lavi et al. (2019) argue that the current congestion-dependent fee mechanism will be insufficient as transaction fee start to gain more importance. Basu et al. (2019) show that an auction-based alternative fee mechanism can lead to a more stable and predictable fee market as the user base increases over time.

⁶ <https://www.blockchain.com/charts/transaction-fees-usd?timespan=2years>

The size of a block limits the processing capacity of the Bitcoin network and creates confirmation delay for users. A seemingly straightforward solution to attenuate this “inefficiency” is to enlarge the block size. However, larger block size diminishes a low-willingness-to-wait user’s incentive to pay for faster confirmation. This is because the desired confirmation speed can now be attained at a lower cost. As a result, miners are disincentivized to participate in the system due to a lack of financial incentives. Note that the security of the bitcoin network depends on the existence of a large number of miners. For example, if a single miner organization controls more than 51% of all mining resources, he/she becomes theoretically capable of revising the transaction history and/or preventing new transactions from being confirmed (Bradbury, 2013). A further drawback of larger block size is increased system latency (Babich and Hilary, 2020), leading to significantly longer broadcasting times. In fact, the tradeoffs involved in choosing block size have instigated heated debates between influential software developers in the cryptocurrency ecosystem (Bogart, 2017). As minting revenue shrinks and the reliance on fee revenue increases over time, the concerns for relaxing block size limit will only become more salient. Therefore, we believe Bitcoin will remain a congested transaction processing system and hence understanding the fee-speed relationship has long-term implications. Next, we provide a stylized representation of the transaction confirmation process.

2.3. Transaction Confirmation on the Bitcoin Network

As depicted in Figure 2, consider a transaction A (txA) that arrives at the Bitcoin network at t_1 , after the generation of an arbitrary block, named B_0 . Given its file size, the sender of txA decides on the satoshi-per-byte fee offer, $TxFee$, at t_1 (1 satoshi = 10^{-8} bitcoin). $TimeSinceB_0$ is the time elapsed from the generation of B_0 to the arrival of txA. Suppose this transaction gets confirmed and added to the blockchain inside block B_N at t_2 . It follows that the difference between t_2 and t_1 is the confirmation time ($ConfTime$) of txA.

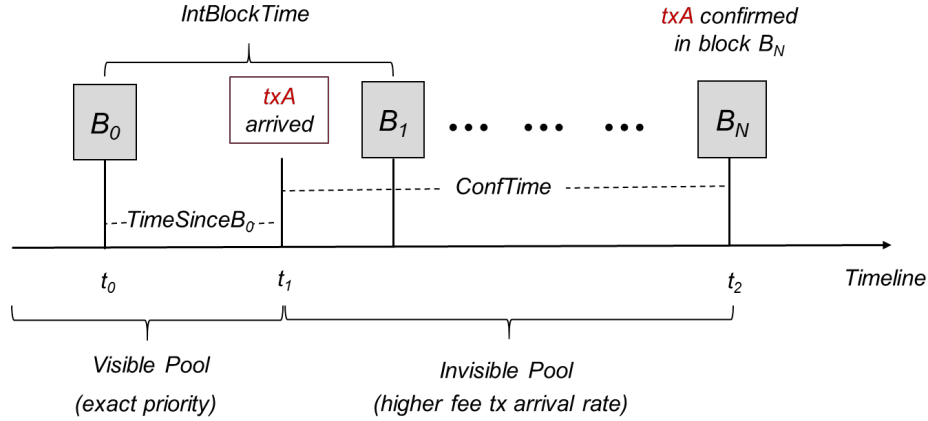


Figure 2. Confirmation Process of an Illustrative Transaction

We first comment on the discoverability of txA to a potential miner. Since miners restart their mining process after the generation of a new block, txA is discoverable to miners of B_2 to B_N . For miners of B_1 , txA is discoverable to a large extent, although not guaranteed. This is because once a miner initiates the mining process, she will frequently (once every few seconds) come back to the memory pool of pending transactions⁷. As a result, if B_1 is generated immediately after t_1 , txA might not be discoverable. Given the rarity of such a situation, we assume full discoverability of txA in the rest of the paper.

The confirmation of txA at t_2 is in fact the joint occurrence of two events: B_N arrives at t_2 and txA is included in B_N . The random process underlying the former is the distribution of IBT , while that of the latter is the probability distribution of txA being included in each of the future blocks. Not-yet-confirmed transactions arrived prior to t_1 form the *Visible Pool*, since they are theoretically observable to the sender of txA . New transactions arriving between t_1 and t_2 are unobservable when $TxFee$ is determined and hence form the *Invisible Pool*.

3. Theoretical Framework and Hypotheses

In this section, we propose a theoretical framework, depicted in Figure 3, that maps the causal path of how the transaction fees chosen by users ($TxFee$) affect the confirmation speed they receive ($ConfTime$).

⁷ This happens whenever she updates the block template. See <https://bitcoin.org/en/developer-guide#solo-mining> for details.

Next, we develop hypotheses that manifest two prominent characteristics of the fee-speed relationship: congestion dependence and tail shrinkage.

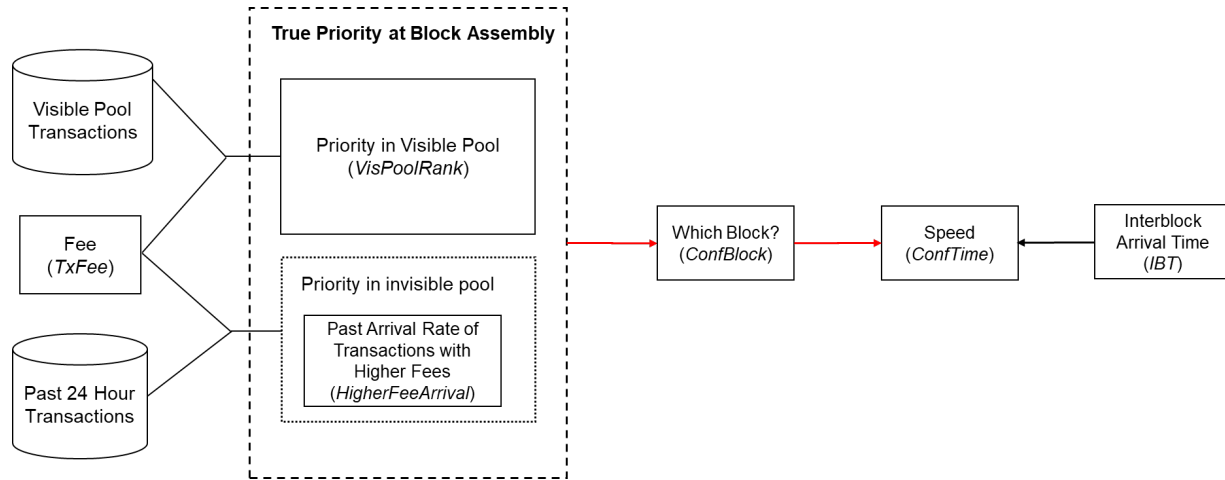


Figure 3. Theoretical Framework of Fee-Speed Relationship

Recall from earlier that *ConfTime* is jointly determined by two underlying factors – the inter-arrival time between blocks (*IBT*) and block of confirmation (*ConfBlock*). Also note that due to the randomness in hashing and the fee's independence from the propagation time, *IBT* can be considered exogenous to fee's effect on confirmation time⁸. These observations imply that *TxFee*'s impact must be transmitted through the block of confirmation. As a result, our framework focuses on the mechanisms that give rise to *TxFee* → *ConfBlock* causal path (highlighted in red in Figure 3). Specifically, an increase in the satoshi-per-byte fee alters the transaction's true priority (big dashed-line box) when a future block is assembled by a miner who would like to maximize his/her rewards by giving priority to higher fee transactions. Unfortunately, this future priority cannot be exactly calculated at the time when *TxFee* is determined. Yet, empowered by the contextual knowledge of the transaction confirmation process (i.e., Section 2.3), we propose two proxy variables that are predictive of the unseen true priority.

⁸ This assumption is specifically suitable at the individual transaction level, which is the level of granularity followed in this study. While it is possible for *IBT* to be affected by a change in average transaction fees at the aggregate level between two proof-of-work difficulty updates (Ilk et al. 2021), it is highly unlikely for an individual transaction fee to instantaneously influence miners' decision and ability to participate to (withdraw from) the network, thereby affecting the aggregate hashing power in the system.

Specifically, true priority is determined by the focal transaction's rank among unconfirmed transactions that arrived ahead of it (i.e., priority in the visible pool) and its rank among future transactions that continue to arrive until the next block is assembled (i.e., priority in the invisible pool). The former, *VisPoolRank*, can be calculated **exactly** from the memory pool at t_l ⁹. Since the latter is unobservable (small dashed-line box), we utilize the concept of effective arrival rate from the queueing literature (Guo and Zipkin, 2007; Mendelson and Whang, 1990) to develop a proxy. Note that only future arrivals with a higher satoshi-per-byte fee would alter focal transaction's rank. As a result, the higher the arrival rate of such transactions, the lower the true rank. We approximate this "effective arrival rate" by the ten-minute average arrival rate of higher fee transactions in the 24 hours prior to t_l , *HigherFeeArrival*. We choose ten minutes as the time interval since this is roughly the mean of *IBT*. While we expect *TxFee*'s impact on *ConfBlock* to be explained by *VisPoolRank* and *HigherFeeArrival*, their ability to approximate the true rank is imperfect, giving rise to an unexplained portion of *TxFee*'s impact. Therefore, we consider all three variables in the main analysis of *ConfBlock*.

3.1. Congestion Dependence

It is a common assumption in the congested-service pricing literature that the extra fee tends to benefit customers the most when the level of congestion is high (Feldman et al., 2017; Van Mieghem, 2000). Consider the CLEAR airport service again. When the regular waiting line is full, paying for CLEAR brings the highest time savings. Since the Bitcoin transaction confirmation system is also a congested service system, sharing similarities with other well-studied systems (reviewed in Section 1), it is expected that the fee-speed relationship in Bitcoin should also exhibit this congestion dependence characteristic. Thus, without the guidance of our theoretical framework, prior literature offers a "black box" approach to capture congestion dependence, which we formalize via H0:

⁹ Since *VisPoolRank* arises deterministically from *TxFee*, we do not use an arrow to connect *TxFee* with *VisPoolRank*. Similarly, the *TxFee*-to-*HigherFeeArrival* relationship is also deterministic and hence no arrow is used. Thus, it is important to not mistaken *TxFee*'s impact on *ConfBlock* depicted in Figure 3 as a typical mediation model where *VisPoolRank* and *HigherFeeArrival* are two mediators. Specifically, no statistical analysis is required for *VisPoolRank* and *HigherFeeArrival* (i.e., the first step in a typical mediation analysis; see Malhotra et al. (2014) for review of the mediation methodology).

H0: *The acceleration effect of transaction fee on confirmation time is larger when the Bitcoin system is more congested.*

We verify this baseline hypothesis through an OLS-based preliminary analysis in Section 5. While this analysis helps to establish the existence of congestion dependence, it does not perform well in demonstrating the magnitude, providing additional motivation for a more context-based approach (Schmenner et al., 2009) to capture the influence of congestion. Our theoretical framework serves exactly this purpose – i.e., it unpacks congestion dependence black box via *VisPoolRank* and *HigherFeeArrival*. Conceptually, congestion level is defined by the system load factor, i.e., demand over capacity (Cachon and Terwiesch, 2008). Given the fixed capacity of each block, congestion in Bitcoin boils down to demand (i.e., arrival rate) – analogous to how congestion is typically measured in healthcare settings (Chan et al., 2017). For the same increase in *TxFee*, the higher the pending demand at t_1 , the larger *TxFee*'s impact on *VisPoolRank*. Similarly, with higher future demand after t_1 , the same increase in *TxFee* is expected to result in a larger reduction in *HigherFeeArrival*. Therefore, our theoretical framework in Figure 3 captures the underlying mechanisms through which congestion affects the fee-speed relationship. This deviates substantially from the straightforward fee-congestion product-term approach implied by H0. We formalize these mechanisms through a pair of hypotheses as follows:

H1a: *A transaction's priority in the visible pool accelerates its confirmation time.*

H1b: *A transaction's priority in the invisible pool, as proxied by the past arrival rate of higher fee transactions, accelerates its confirmation time.*

3.2. Tail Shrinkage

Discussions among Bitcoin users reveal that for an average user who can tolerate a reasonable amount of waiting, the primary motivation for paying transaction fee is to avoid being “forgotten in the wild”¹⁰. This refers to the situation where a transaction waits a long period before being added to a block. Our data analysis reveals this situation indeed occurs at a non-trivial frequency – i.e., we find 2.3% transactions in

¹⁰ <https://cointelegraph.com/news/why-is-my-bitcoin-transaction-taking-so-long-heres-why>;
<https://bitpay.com/blog/transaction-delays/>

our data set to have experienced a confirmation time longer than a day. Thus, from a user's perspective, insights regarding the fee-speed relationship should not be limited to the mean of the confirmation time distribution, but rather extend to the shape of this distribution. Of particular importance are the higher quantiles (i.e., tail of the distribution), since long waiting times are shown to cause lower service reuse (De Vries et al., 2018) and customer satisfaction (Goes et al., 2018). Extending the scope of our analysis to the tail of the waiting time distribution is also in line with the typical way firms communicate waiting time guarantees to customers (Kumar et al., 1997): "your wait is 10 minutes" is commonly perceived by customers as "I (most likely) need to wait no more than 10 minutes", rather than "my average waiting time is 10 minutes". Similarly, waiting time management in practice often focuses on a target service level rather than the expected waiting time, such as the widely used 80/20 service level (80% of the calls need to be answered within 20 seconds) in the call center industry (Whitt, 2005).

Grounded in the Bitcoin transaction confirmation process described in Section 2.3, we theorize in the following why fee's impact might be larger on the tail of confirmation time distribution. First, consider a simplified scenario where a transaction's probability of being included in the first available block is p . In addition, if the transaction is not included in the previous block, the probability for its inclusion in the next block remains as p . It follows that the probability density function for block of confirmation in this scenario is characterized by a negative binomial distribution: $\Pr(n) = (1 - p)^{n-1}p$, which is known to be skewed to the right (i.e., long right tail) with a variance of $\frac{1-p}{p^2}$. Offering a higher transaction fee increases p and hence shrinks the variance. Given the right skewness (i.e., more variance comes from the tail), the marginal impact of a fee increase is expected to be more pronounced on the tail of confirmation time distribution. Interestingly, this asymmetric impact of fee on mean versus tail is similar to the impact of adding more service agents on customer call center waiting time – a classic congested service setting in OM (Ilk et al., 2020; Whitt, 2005). Second, small fee creates the danger of cascading the focal transaction in the rank as more high fee future transactions arrive to the memory pool. As a result, conditional on not being included in the previous block, the probability of inclusion in the next block might be even lower (smaller than p).

This implies the long tail problem might be particularly strong when fee is low. The following hypothesis formalizes the tail shrinkage effect of transaction fee:

H2: *Transaction fee exerts a larger impact on the higher quantiles of the confirmation time distribution.*

The preliminary analysis in Section 5 provides reduced-form evidence for the existence of tail shrinkage. Our framework-based analysis in Section 6 demonstrates structural mechanisms behind this phenomenon since it models *TxFee*'s impact on *ConfBlock* via a block-by-block approach. Although fee's impact on confirmation time is only transmitted through the block of confirmation, for completeness, we propose an efficient method to compute the distribution of confirmation time by convoluting *ConfBlock* with *IBT*. We formally test H1 and H2 in Section 7 after the introduction of this computational method. We also discuss its practical use for fee recommendation.

4. Data and Variables

4.1. Data Collection

We collect transaction and block data from www.blockchain.com – i.e., a blockchain explorer platform launched in 2011. A major advantage of this website is that it records not only the timestamp of a transaction's inclusion in the blockchain ledger (t_2 in Figure 2), but also the timestamp that it was first seen in the network (t_1), which allows us to approximate the confirmation time. We consider the 4-month time period from November 11th 2017 to March 10th 2018 for two reasons. First and foremost, the Bitcoin system was frequently congested (daily load factor frequently above 1) and the users were paying a wide range of transaction fees during this period, leading to high volatility of confirmation times. This allows us to estimate our framework under varying system circumstance. Second, while the Bitcoin protocol updates frequently, it was under the same version (v0.16) during the entirety of our data period.

Transactions in the raw data is organized by blocks – a common feature of Blockchain data. The 4-month period has a total of 18,953 blocks and more than 33 million transactions. Each block is stored in a JSON file, which we access by interacting with the API of www.blockchain.com. While some variables of interest can be readily extracted from the transaction-level information available in the raw data such as

ConfTime and *TxFee*, some others require a computationally intensive construction procedure. Take *VisPoolRank* for example. To calculate this variable for txA (see Figure 2), we need to first compile a list of transactions arrived before yet confirmed after t_l – i.e., construct the memory pool at t_l . Given the long tail of *ConfTime* (maximum of 43,012.83 minutes or 30 days), the memory pool reconstruction necessitates checking all transactions confirmed within 30 days of t_l . Thus, we augment our data collection time frame to 30 days before November 11, 2017 and 30 days after March 10, 2018 to accommodate the above variable construction requirement. Once t_l 's memory pool is constructed, *VisPoolRank* is simply the position of txA after ranking all transactions by satoshi-per-byte fee.

The memory pool is dynamically updated as new transactions arrive and pending transactions are confirmed. Thus, processing the entire raw data requires constructing memory pool iteratively for each of the 33 million transactions, which imposes a formidable computational burden¹¹. As a result, we sample 2400 blocks (i.e., 20 blocks per day on average) and perform variable construction for all transactions from these blocks, yielding over 4 million observations. To optimize the data preparation process, we utilize batch processing and multithreading. The former groups transactions with the same timestamp in a batch to reduce potential duplicated computation, while the latter runs the software program in multiple parallel threads. Our resulting run time for variable construction is roughly two weeks.

¹¹ It takes approximately 1 second for a relatively powerful server (with Intel® Xeon® Processor E5-2650 v4 and 16 Gigabytes memory) to process one target transaction and create all variables used in this study. This amounts to over 1 year for all 33 million transactions in the absence of hardware and software failures.

Table 2: Variable Definitions and Summary Statistics (N = 4,282,594)

Variable	Definition	Mean	St. Dev.	Min	Pctl (25)	Pctl (75)	Max
<i>ConfTime</i>	Time elapsed between arrival and confirmation	183.28	1,386.47	0	3.35	21.07	43,012.83
<i>ConfBlock</i>	Order of the future block where the transaction is confirmed in	19.5	151.76	1	1	2	4,864
<i>TxFee</i>	Transaction fee in satoshi per byte	295.79	295.69	0	106.44	436.56	54,537.31
<i>VisPoolRank</i>	<i>TxFee</i> rank of a transaction within the visible pool	2,583.06	7,845.44	1	270	2,013	173,921
<i>VisPoolUptoSize</i>	Total file size of transactions in the visible pool with a higher <i>TxFee</i> (in megabytes)	1.56	6.6	0.0002	0.12	0.86	205.14
<i>HigherFeeArrival</i>	Ten-minute average arrival rate of transactions with a higher <i>TxFee</i> in the past 24 hours	968.42	682.34	0	403.21	1403.38	3524.83
<i>VisPoolNumTx</i>	Number of transactions in the visible pool	40,819.58	40,795.54	0	8,338	57,672	182,218
<i>VisPoolAvgFee</i>	Average <i>TxFee</i> for transactions in the visible pool	79.69	60.19	0.9	31.51	110.89	320.17
<i>PastAvgFeeBlock1</i>	Within the past 6 blocks, average <i>TxFee</i> for transactions with <i>ConfBlock</i> = 1, 2, 3, 4, 5, 6	337.04	230.18	22.7	158.87	477.54	1,222.04
<i>PastAvgFeeBlock2</i>		244.89	206.46	2.39	83.36	372.43	1,049.71
<i>PastAvgFeeBlock3</i>		190.37	198.73	0.48	25.7	311.33	3,166.86
<i>PastAvgFeeBlock4</i>		170.17	190.31	0.48	19.25	256.82	13,192.57
<i>PastAvgFeeBlock5</i>		159.32	187.27	0.6	15.24	240.73	2,875.85
<i>PastAvgFeeBlock6</i>		147.89	169.74	0.56	15.19	220.52	4,380.71
<i>TimeSinceB₀</i>	Time elapsed between latest block and arrival	8.13	8.45	0.02	2.3	11.1	95.1
<i>TxSize</i>	File size of a transaction in kilobytes	0.58	2.78	0.16	0.22	0.37	203.63
<i>ExchangeRate</i>	Bitcoin price in 1000 USD	11.96	3.31	5.5	9.36	14.76	19.64
<i>NextBlockRank</i>	<i>TxFee</i> rank of a transaction at next block	3,453.34	8,141.13	2	650	3,310	175,936
<i>TxValue</i>	Transaction value in bitcoins	7.6	123.62	0	0.01	0.88	51,044.08
<i>TxValueVPR</i>	<i>TxValue</i> rank of a transaction within the visible pool	13,652.09	21,215.98	1	1,642	16,374	179,824
<i>TxValueHFA</i>	Ten-minute average arrival rate of transactions with a higher <i>TxValue</i> in the past 24 hours	1,040.68	696.19	0	474.60	1,491.40	3,520.16

All time measures are in minutes.

4.2. Data Summary

Table 2 presents variable definition and summary statistics. The correlation matrix is provided in Appendix A. Apart from the variables discussed previously in Figures 2 and 3, we further construct *VisPoolNumTx*, *VisPoolAvgFee*, the set of *PastAvgFeeBlock* variables, and *ExchangeRate* as control variables. In addition, the last three variables related to transaction value are used as instrumental variables to address potential endogeneity concerns of *TxFee* (see Section 6.2). Last, we control for *TxSize* so that *TxFee*'s impact can be interpreted in isolation of file size.

The fact that *ConfTime* has a long tail – discussed in Section 3.2 and shown in Figure 4a – implies that *ConfBlock* should exhibit a similar characteristic, which is demonstrated in Figure 4b. Specifically, confirmation in the first ten blocks makes up 93.2% of the sample, with the first three blocks highly densely populated. *ConfBlock* between 11 and 500 together makes up another 6.2%. The histogram in Figure 4a plots the distribution of *ConfBlock* in this segment. The leftover cases correspond to *Confblock* > 500, contributing 0.6%. These descriptive characteristics of *ConfBlock* are especially informative for model specification in our main analysis.

On the other hand, Figure 4b shows that *TxFee* spreads over the range of 0 to 800 satoshi per byte. This suggests that customers' willingness to pay is highly heterogeneous with a sizeable fraction of paying a significant amount of fees. This level of fee magnitude and heterogeneity is very different from a few years ago. Möser and Böhme (2015) report that between 2011 and 2012, majority of the bitcoin transactions do not offer any fee (i.e. $TxFee = 0$). Another feature of *TxFee*'s distribution is its bumpiness. That is, there are many spikes – *TxFee* values with high frequency (top five spikes marked in Figure 4b). We revisit this observation in Section 6.2 for its insights on the strength of *TxFee* endogeneity.

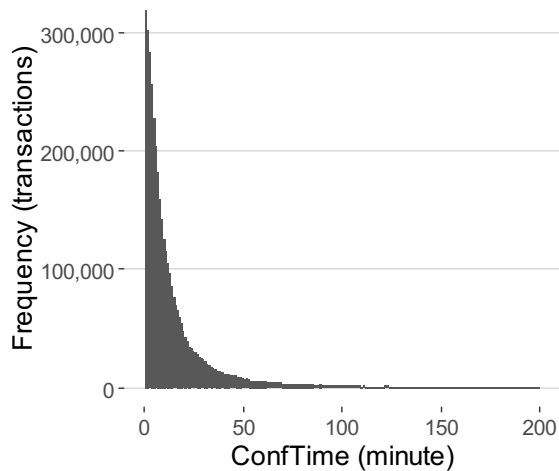


Figure 4a. Histogram of *ConfTime*¹²

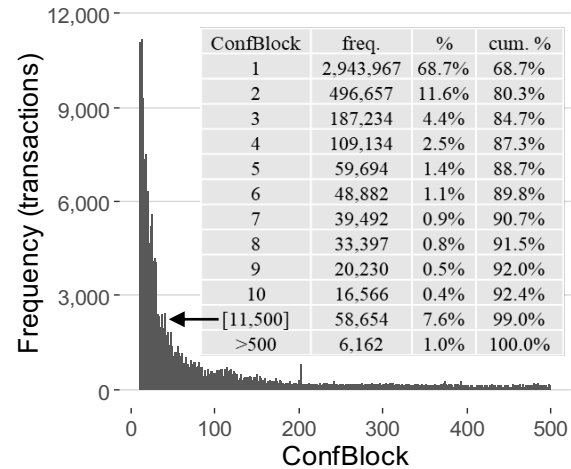


Figure 4b. Histogram of *ConfBlock*

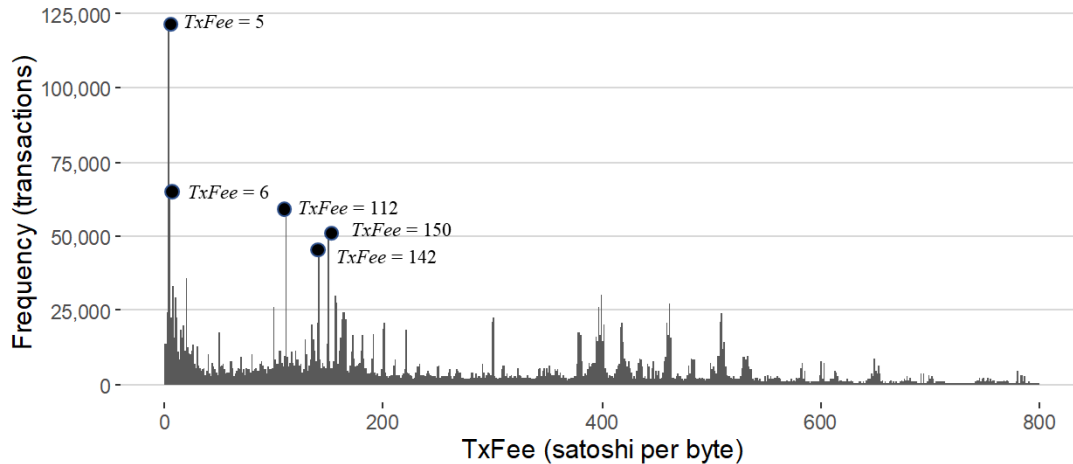


Figure 4c. Histogram of *TxFee*¹³

5. Preliminary Analysis

Since running regressions on the entire dataset is prohibited by computer memory limit, we randomly split the data into two equal-sized subsamples. All analyses are run on the first subsample. Replication results using the second subsample is discussed in Section 6.5, which is almost identical as one would expect from our large sample size.

¹² *ConfTime* > 200 accounts for 5.5%.

¹³ *TxFee* > 800 accounts for 5.4%.

In this section, we consider two sets of preliminary analyses, starting with an OLS-based analysis for verifying the baseline hypothesis H0, as discussed in Section 3.1. The simplest approach to study the fee-speed relationship is to regress *ConfTime* on *TxFee* via an OLS estimator. Given the exponential-like shape of the *ConfTime* distribution (see Figure 4a), we use its logged version as the dependent variable. H0 implies a product-term approach to test how congestion alters the fee-speed relationship. The “real-time” congestion faced by the focal transaction is captured by *VisPoolNumTx*. Thus, our first preliminary analysis has the following specification:

$$ConfTime_log_i = \alpha^{OLS} + \beta^{OLS} X_i + \Gamma^{OLS} Controls_i + \varepsilon_i \quad (1)$$

X_i contains our independent variables of interest: *TxFee* and *VisPoolNumTx*. Table 3 presents the key results. The exact list of control variables and the full regression results are shown in Appendix B.

Table 3: Preliminary Analysis: OLS Regression Results

	Model OLS1	Model OLS2
<i>TxFee</i> ($\times 10^{-2}$)	-0.176*** (0.0005)	-0.181*** (0.0007)
<i>VisPoolNumTx</i> ($\times 10^{-3}$)	0.004*** (0.0001)	0.004*** (0.0001)
<i>TxFee_x_VisPoolNumTx</i> ($\times 10^{-5}$)		-0.00006*** (0.000007)
<i>Controls</i>	see Appendix B	see Appendix B
R ²	11.1%	11.1%

***p<0.001. $\times 10^{-n}$ means the coefficient is scaled up by a factor of 10^n for exposition purpose.

As expected, Model OLS1 shows that higher fees reduce confirmation time, whereas higher congestion leads to longer confirmation time. Model OLS2 adds the interaction term between *TxFee* and *VisPoolNumTx*. Its negative direction and statistical significance offer support for H0: when the number of pending transactions in the memory pool grows (i.e. more congested), the speed acceleration effect of fee is stronger. However, the magnitude of congestion dependence captured by this simple product-term approach (-6×10^{-10}) is miniscule, providing evidence that a more sophisticated, context-based approach is needed to capture how congestion affects the fee-speed relationship.

The second preliminary analysis utilizes quantile regression to investigate how the impact of fee differs across different centiles of the confirmation time distribution, as discussed in Section 3.2. Using the same

variable specification as in Model OLS1, the standard quantile regression equation (Koenker and Bassett Jr, 1978) is as follows:

$$Q_{\tau}(ConfTime_{log_i}) = \alpha_{\tau}^{QR} + B_{\tau}^{QR} X_i + \Gamma_{\tau}^{QR} Controls_i \quad (2)$$

where τ specifies the centile of interest. For example, when $\tau = 0.5$, the above equation estimates the medium of *ConfTime_log* as a function of *TxFee*, *VisPoolNumTx*, and controls. We vary τ from 0.1 to 0.9 with increments of 0.1 (i.e. 1st to 9th decile) to examine the impact of fee on a range of locations on the confirmation time distribution¹⁴.

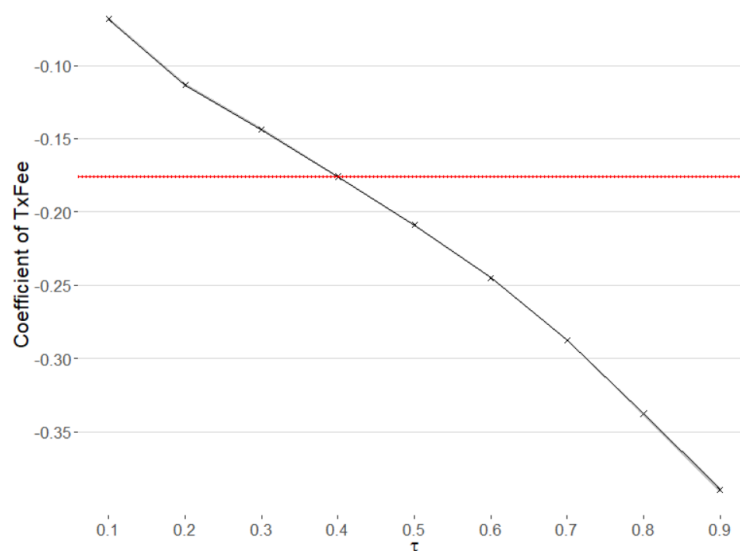


Figure 5. Effect of *TxFee* Estimated with Quantile Regression¹⁵

While full regression results are relegated to Appendix B, we plot in Figure 5 the coefficient of *TxFee* against τ (always significant at $p < 0.001$ level), which constitutes an intuitive visual illustration of how the impact of fee becomes larger (i.e., more negative) as we move up the deciles of the confirmation time distribution. At the two extremes (1st and 9th deciles), fee's impacts are drastically different – the latter is about 6 times stronger than the former (-0.068 vs -0.39). Another interesting comparison to make is between

¹⁴ Due to the large size of our sample, we implement the Frisch–Newton interior point method for faster estimation (Portnoy and Koenker 1997) and calculate standard errors with the asymptotic covariance matrix method (Koenker and Bassett Jr 1978).

¹⁵ The confidence interval is plotted, albeit not visible in the figure due to its tightness.

fee's impact on the mean (estimated in Model OLS1; red horizontal line in Figure 5) and the 9th decile ($\tau = 0.9$) of *ConfTime_log*. As discussed in Section 3.2, customers might care more about the tail of confirmation time distribution than its mean. The above result shows that a higher fee is much more effective at improving the 90% service level (effect size at $\tau = 0.9$: -0.344) than the mean time (effect size from OLS1: -0.115). We revisit this tail shrinkage effect in Section 7 using our framework-based estimation results.

6. Framework-Based Analysis

6.1. Model Specification

The data generating process underlying *ConfBlock* resembles a sequence of binary-choice decisions. That is, *ConfBlock* = n is in fact the outcome of n miner decisions (i.e., not included in $B_1, B_2, \dots, B_{n-1}$; included in B_n). This calls for a sequential logit approach (SL) with one logit regression estimating the inclusion probability in each block conditional on the transaction not being included up to the previous block (Tutz, 1991). This approach is frequently used to study subjects' transition from one step to the next, such as students' primary, secondary, and tertiary educational attainment (Buis, 2017). To estimate the impact of *TxFee* along with *VisPoolRank* and *HigherFeeArrival*, in the most delicate manner, it appears that we should apply the SL model to the entire range of *ConfBlock*. This is, however, unrealistic due to the large variation and long tail of *ConfBlock* as shown in Figure 4a, and also unnecessary since majority of the transactions are confirmed in the first few blocks. As a result, we use a two-part modeling approach – a transaction's inclusion in the first three blocks is estimated with a three-step SL model (84.7% of total sample), and in the 4th or later blocks are modeled through linear regression with logged *ConfBlock* as the dependent variable (15.3%). To distinguish the latter part from the OLS preliminary analysis in Section 5.1, we label it as LN (i.e., Log Normal). The two-part model is given as follows:

$$I_j(\alpha^{SLj} + B^{SLj}X_i + \Gamma^{SLj}Controls_i + \epsilon_i > 0) \text{ where } j = 1,2,3 \quad (3a)$$

$$ConfBlock_log_i = \alpha^{LN} + B^{LN}X_i + \Gamma^{LN}Controls_i + \epsilon_i \quad (3b)$$

Eq.(3a) corresponds to the SL part of the model, where I_j is an indicator variable equal to 1 if *ConfBlock* = j and to 0 if *ConfBlock* > j (i.e., a transaction can only be confirmed later if not confirmed in the j th block). Eq.(4a) is estimated three times for $j = 1,2,3$. Each estimation sample consists of transactions with

$ConfBlock \geq j$. Eq.(3b) corresponds to the LN part of the model. This estimation sample consists of transactions with $ConfBlock \geq 4$. For both Eq. (3a) and (3b), X_i includes the three key independent variables of interest: $TxFee$, $VisPoolRank$, and $HigherFeeArrival$. We provide a detailed discussion regarding the source of $TxFee$ endogeneity and our solution strategy in the next section.

6.2. Transaction Fee Endogeneity: Source and Solution Strategy

Source of Endogeneity. Strategic customer behavior is documented in OM settings such as waiting for a potential price drop of flight tickets (Li et al., 2014) and restraining from booking a premium hotel room to be targeted a standby upgrade offer (Yilmaz et al., 2019). It is reasonable to expect that when faced with system congestion and empowered by Name-Your-Own-Price, Bitcoin users would strategically choose a $TxFee$ in hope to achieve a desired confirmation speed. In traditional NYOP settings, it is optimal for the customer to name a price that discounts her true willingness-to-pay. The discount factor is a function of her hassle cost for making repeated bids (Terwiesch et al. 2005, p.349). A Bitcoin user's willingness-to-pay for $TxFee$ is, however, pegged with confirmation speed; and the pegging mechanism is ambiguous. As a result, Bitcoin users might rely on readily available fee recommendation services, such as the Bitcoin Core Wallet¹⁶, or collect their own congestion signals to set $TxFee$, depending on their level of sophistication. Baur et al. (2018) present a categorization of Bitcoin users: active investor, passive investor, hybrid, currency user, miner, and tester. Ilk et al. (2021) find evidence in line with cross-user-type behavioral differences. In terms of choosing $TxFee$, different user types will conceivably have varying degrees of sophistication. To summarize, $TxFee$ endogeneity in identifying the causal fee-speed relationship arises from congestion-related omitted variable bias, i.e., we cannot fully observe all congestion signals used by all user types to set fees.

Descriptive Evidence of User Sophistication. How sophisticated are users when reacting to changes in congestion level? The summary statistics of $TxFee$ offer some insights. The bumpiness of the histogram in Figure 4c implies that at least a nontrivial fraction of users are not likely to be strategic in choosing their

¹⁶ <https://gist.github.com/morcos/d3637f015bc4e607e1fd10d8351e9f41>

fees. This is because congestion at the transaction-level changes continuously. If *TxFee* is highly responsive to congestion signals, its distribution should be much smoother than the pattern shown in Figure 4c. In the following, we propose a two-fold solution strategy for addressing *TxFee* endogeneity.

Congestion Signal Controls. Part I of our strategy directly controls for the most visible congestion signals, as well as signals derived from the Bitcoin Core Wallet, due to its popularity among less sophisticated users, who are most likely those in the passive investor, hybrid, and currency user categories (Wilson, 2020). First, we have a measure of congestion, *VisPoolNumTx*, that is highly granular and clocked at the exact time when the fee decision is made (i.e., t_l in Figure 2). Second, users might base their fee selection on what was offered in the current memory pool. To this end, we calculate the average fee level among pending transactions at t_l , *VisPoolAvgFee*. Third, Bitcoin Core Wallet uses a simple approach to suggest fees: calculating stratified averages of what was offered in the most recent 6 blocks. Using the same logic, we construct a set of *PastAvgFeeBlockN* variables, each capturing average *TxFee* for transactions with *ConfBlock* = 1, 2, 3, 4, 5, 6. Fourth, the *ExchangeRate* (i.e., 1 bitcoin can be exchanged for how many US dollars) is a primary driver for bitcoin trading activities and hence induces congestion as it fluctuates (Ilk et al., 2021). As some users might consider *ExchangeRate* when setting *TxFee*, we also control for *ExchangeRate*¹⁷. The control variable approach discussed above is especially advantageous when the mechanism through which the endogeneity arises is clear and proxy variables for this mechanism are available (Card, 1999). Indeed, Rossi (2014) recommends “*measurement of the unobservables which create the endogeneity concerns in the first place*”, in line with conclusions from recent methodology reviews (Ketokivi and McIntosh, 2017; Lu et al., 2018).

Instrumental Variables. Part II of our strategy uses instrumental variables to further addresses the leftover omitted variable bias arising from sophisticated users who consider congestion signals outside our set of controls. Our key idea for searching valid instruments centers around our contextual knowledge that *TxValue* – the bitcoin value of the transaction (exclude fee) – should drive the fee choice but is otherwise

¹⁷ We thank an anonymous reviewer for this suggestion.

unrelated to confirmation speed. Specifically, users sending a higher value transaction are likely to be willing to pay a higher transaction fee, giving rise to a theoretical rationale for *TxValue*'s effect on *TxFee* and hence satisfying the *relevance* condition. For the *exogeneity* condition, recall from Section 2.2 that a miner has two revenue sources, none of which are related to *TxValue*. Hence, it cannot directly affect the block of confirmation and confirmation time.

As *VisPoolRank* and *HigherFeeArrival* are partially determined by *TxFee*, they are in principle endogenous too¹⁸. However, we expect them to be much less correlated with the source of endogeneity and hence suffer endogeneity bias to a lesser magnitude than *TxFee* is. This is because, as shown by Ilk et al. (2021), the congestion-to-fee effect is system-wide: the congestion signal picked up by the focal transaction is likely highly correlated with that seen by its adjacent preceding transactions (i.e., transactions in the t_l memory pool and arrived within 24 hours prior to t_l , respectively). As a result, while it is easy for the focal transaction to respond to congestion with a higher fee, it is hard to “climb” the priority ladder when the congestion response is systematic. Nonetheless, we construct *TxValue* analogs of *VisPoolRank* and *HigherFeeArrival* as additional instruments, labeled *TxValueVPR* and *TxValueHFA*. The rationale is straightforward: if *TxValue* is correlated with *TxFee*, the focal transaction's value rank in t_l memory pool and past 24-hour arrivals should predict its fee rank.

Following the optimal instrument literature (Newey, 1990) and common practice (Angrist and Keueger, 1991), we consider higher order terms as further instruments so that over-identification tests can be performed. Specifically, as the bitcoin value and the rank of this value might show complementarity or substitutability in affecting fee choice (i.e., user gives more/less attention to *TxValue* depending on other's *TxValue*), we use the *TxValue_x_TxValueVPR* and *TxValue_x_TxValueHFA* interactions as additional instruments. To summarize, five instruments are used for three potentially endogenous regressors.

6.3. Assessing Instrument Validity

¹⁸ We thank an anonymous reviewer for pointing out this possibility and encouraging us to search additional instruments.

Given the availability of the full array of tests for endogeneity and instrument quality in the linear regression context, we use the log-normal model in Eq. (3b) for conducting these tests (fitted to all *ConfBlock* values, including $j \leq 3$). The guidance provided in Lu et al. (2018, Figure 2) is strictly followed. The first four columns of Table 4 shows results from the two-stage-least-squares (2SLS) estimator (i.e., one first stage for each endogenous regressor). Cluster-robust standard errors (clustered by block) are used to reflect the fact that transactions in the same block are likely correlated.

Table 4: Assessing Endogeneity and Instrument Validity (N = 2,141,297)

DV	2SLS				LIML	GMM	OLS
	1st Stage <i>TxFee</i>	1st Stage <i>VisPoolRank</i>	1st Stage <i>HigherFeeArrival</i>	2nd Stage <i>ConfBlock</i>	2nd Stage <i>ConfBlock</i>	2nd Stage <i>ConfBlock</i>	<i>ConfBlock</i>
<i>TxValue</i>	0.037† (0.020)	0.016 (0.073)	0.0079 (0.0089)				
<i>TxValueVPR</i> $\times 10^{-3}$	-0.0035** (0.0013)	0.063*** (0.0093)	-0.0042*** (0.00058)				
<i>TxValueHFA</i> $\times 10^{-3}$	-0.21*** (0.026)	-0.20 (0.13)	0.26*** (0.015)				
<i>TxValue_x_TxValueVPR</i> $\times 10^{-3}$	-0.0049* (0.0019)	0.021*** (0.0064)	-0.0020** (0.00063)				
<i>TxValue_x_TxValueHFA</i> $\times 10^{-3}$	0.099*** (0.017)	-0.27*** (0.045)	0.034*** (0.0089)				
<i>TxFee</i> $\times 10^{-2}$				-0.39*** (0.069)	-0.42*** (0.079)	-0.24*** (0.061)	-0.018*** (0.0036)
<i>VisPoolRank</i> $\times 10^{-3}$				0.082*** (0.014)	0.078*** (0.015)	0.10*** (0.013)	0.064*** (0.0023)
<i>HigherFeeArrival</i> $\times 10^{-3}$				0.63*** (0.10)	0.60*** (0.11)	0.78*** (0.095)	0.62*** (0.024)
Weak instrument tests							
Kleibergen and Paap (2006) Rank Lagrange Multiplier test				505.74(3)***			
Sanderson-Windmeijer (2016) Chi-squared test				448.41(3)***			
Sanderson-Windmeijer (2016) F test				89.64***			
Endogeneity test							
GMM distance test				409.38(3)***			
Over-identification test							
Difference-in-Sargan C statistic for <i>TxValue</i>				0.03(1)	0.01(1)	0.03(1)	
Difference-in-Sargan C statistic for <i>TxValueVPR</i>				2.41(1)	2.56(1)	2.41(1)	
Difference-in-Sargan C statistic for <i>TxValueHFA</i>				3.51(1)	3.08(1)	3.51 (1)	

A full set of control variables (same as those shown in Table 5) is included in all regressions for estimation yet omitted from the table for brevity.

†p<0.1, *p<0.05, **p<0.01, ***p<0.001; cluster robust standard errors in parenthesis, clustered by block ID.

LIML and GMM estimators share the same first stage regressions, weak instrument test, and endogeneity test results as 2SLS, which are omitted for brevity.

$\times 10^{-n}$ means the coefficient is scaled up by a factor of 10^n for exposition purpose.

Expected IV Effect. We have a clean *a priori* expectation for the main effects of the instruments in the first stage. That is, *TxValue*, *TxValueVPR*, and *TxValueHFA* should each have a positive effect on *TxFee*, *VisiblePoolRank*, and *HigherFeeArrival*, which is confirmed by first stage coefficients highlighted in bold.

IV Relevance. Most of the instrumental variable coefficients register a significant effect. As recommended by Lu et al. (2018, p.60) for the context of heteroskedastic standard errors, we perform the Kleibergen and Paap (2006) Lagrange Multiplier test, Sanderson-Windmeijer (2016) Chi-squared test, and F test, all rejecting the null hypothesis (model is under-identified or weakly identified) at the 0.001 level.

*IV Exogeneity*¹⁹. We test overidentifying restrictions using the Difference-in-Sargan C statistic (Hayashi, 2011) for each of the three instruments, none of which is statistically significant. This is consistent with the null that each of the instrument is exogenous assuming other instruments are exogenous.

Triangulation. We triangulate our 2SLS results by refitting the log-normal model with limited information maximum likelihood (LIML) and generalized method of moments (GMM) estimators. Their second stage results are shown in Table 4. While the coefficients on the three endogenous regressors do exhibit some variation across estimators, such variation is not statistically significant, which can be intuitively seen from the size of the standard errors.

Magnitude of Endogeneity. Addressing endogeneity via instrumental variables causes loss of efficiency which can only be justified if the simpler no-instrument model is sufficiently biased (Hausman, 1978). We perform an extension of the popular Durbin-Wu-Hausman test – the GMM distance test, which is robust to the violation of homoskedastic errors (Baum et al., 2003; Hayashi, 2011; Lu et al., 2018). It rejects the null hypothesis at 0.001 level and offers statistical evidence for the existence of endogeneity. As recommended by prior methodological studies (Lu et al 2018, p.59; Rossi 2014), we also present OLS results in the last column and comment on the direction and magnitude of coefficient changes for the endogenous regressors. Given our source of endogeneity discussion in the last section, we can make two *a priori* predictions. First, omitted variable bias works in the opposite direction as *TxFee*'s true effect: unobserved congestion signals create a positive correlation between *TxFee* and *ConfBlock*, while *TxFee*'s causal effect should be negative. Hence, this bias should attenuate *TxFee*'s effect in the OLS regression. Second, the change in *VisPoolRank*

¹⁹ Note that unlike the relevance condition, the exogeneity condition of instrument validity cannot be formally verified by statistical tests (Rossi 2014, p.665). Instead, exogeneity is typically argued through contextual knowledge of the data (Deaton 2010, p.431), which we have provided in the previous section.

and *HigherFeeArrival* coefficients should be smaller since system-wide response to congestion signal makes the endogeneity bias less pronounced. Both predictions are verified by comparing the OLS and 2SLS estimates, extending additional confidence to our endogeneity solution.

6.4. Main Regression Results

Table 5: Block of Confirmation Regression Results

DV Sample	Model SL0 I(CB= 1) All Obs.	Model SL1 I(CB = 1) All Obs.	Model SL2 I(CB = 2) CB > 1	Model SL3 I(CB = 3) CB > 2	Model LN log(CB) CB > 3	Model RNB <i>NextBlockRank</i> All Obs.
<i>Intercept</i>	2.451*** (0.018)	4.085*** (0.028)	0.747*** (0.045)	-0.728*** (0.067)	0.433*** (0.038)	123.156*** (9.741)
<i>TxFee</i> $\times 10^{-2}$	1.777*** (0.010)	0.859*** (0.014)	0.784*** (0.024)	0.505*** (0.035)	-0.287*** (0.023)	-185.294*** (4.821)
<i>VisPoolRank</i> $\times 10^{-3}$		-0.021*** (0.003)	-0.153*** (0.003)	-0.206*** (0.004)	0.054*** (0.002)	1002.775*** (1.077)
<i>HigherFeeArrival</i> $\times 10^{-3}$		-2.105*** (0.031)	-0.586*** (0.038)	-0.13* (0.057)	1.299*** (0.032)	653.875*** (11.124)
<i>VisPoolNumTx</i> $\times 10^{-3}$	-0.003*** (0.000)	-0.007*** (0.000)	-0.006*** (0.000)	0.0004 (0.001)	0.002*** (0.000)	2.805*** (0.080)
<i>VisPoolAvgFee</i> $\times 10^{-2}$	-1.325*** (0.010)	-1.199*** (0.012)	-0.338*** (0.025)	0.028 (0.041)	0.647*** (0.026)	241.399*** (4.281)
<i>PastAvgFeeBlock1</i> $\times 10^{-2}$	-0.94*** (0.008)	-0.376*** (0.011)	-0.17*** (0.009)	0.032*** (0.007)	0.011** (0.004)	61.936*** (3.863)
<i>PastAvgFeeBlock2</i> $\times 10^{-2}$	-0.155*** (0.004)	0.006 (0.004)	-0.142*** (0.007)	-0.017 (0.010)	-0.047*** (0.006)	-15.172*** (1.421)
<i>PastAvgFeeBlock3</i> $\times 10^{-2}$	-0.057*** (0.003)	-0.052*** (0.003)	-0.037*** (0.004)	-0.045*** (0.005)	-0.027*** (0.003)	54.392*** (1.027)
<i>PastAvgFeeBlock4</i> $\times 10^{-2}$	0.039*** (0.003)	-0.023*** (0.003)	0.05*** (0.004)	0.055*** (0.004)	0.054*** (0.002)	-7.7*** (0.073)
<i>PastAvgFeeBlock5</i> $\times 10^{-2}$	-0.06*** (0.003)	-0.037*** (0.003)	-0.074*** (0.004)	-0.222*** (0.006)	0.034*** (0.003)	23.133*** (0.894)
<i>PastAvgFeeBlock6</i> $\times 10^{-2}$	-0.034*** (0.004)	-0.04*** (0.004)	-0.058*** (0.005)	-0.053*** (0.008)	-0.008* (0.004)	1.386 (1.161)
<i>TimeSinceB0</i>	-0.068*** (0.000)	-0.071*** (0.000)	-0.013*** (0.000)	-0.002** (0.001)	-0.005*** (0.000)	-13.293*** (0.119)
<i>TxSize</i>	-0.021*** (0.001)	-0.014*** (0.001)	-0.022*** (0.001)	-0.013*** (0.002)	0.01*** (0.001)	-4.837*** (0.262)
<i>ExchangeRate</i>	-0.075*** (0.001)	0.022*** (0.002)	0.017*** (0.004)	0.075*** (0.006)	-0.041*** (0.004)	0.737 (0.655)
1 st Stage Residuals	included	included	included	included	included	included
Fixed Effects						
<i>Month</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Day of Week</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Hour</i>	Yes	Yes	Yes	Yes	Yes	Yes
N	2,141,297	2,141,297	669,496	421,552	327,580	2,141,297
LL	-960,240	-907,023.00	-361,143.00	-188,853.00		
R ²					37.40%	98.60%

*p<0.05, **p<0.01, ***p<0.001; *ConfBlock* is abbreviated as CB in header.

$\times 10^{-n}$ means the coefficient is scaled up by a factor of 10^n for exposition purpose.

For computational efficiency, one instrument is used in each of the three first stages (e.g., *TxValue* for *TxFee*).

All first stage residuals are significant except for *VisPoolRank* in Model RNB.

Month/Day of Week/Hour contains 4/6/23 dummy variables, respectively, to capture the heterogeneity across months, days of the week and hours of the day.

Having assessed the validity of our instruments, we proceed to estimating the full regression model. To ensure consistency across Eq. (3a) and (3b), we use the control-function approach popular in the context of choice models (Petrin and Train, 2010), while is also unbiased for linear models (Vella and Verbeek 1999). In Table 5, Model SL1-3 present results for Eq. (3a) and Model LN for Eq. (3b). Model SL0 is identical to Model SL1 except for the exclusion of *VisPoolRank* and *HigherFeeArrival* (and hence only *TxValue* is used as instrument in the 1st stage).

We first compare *TxFee* coefficients in Model SL0 vs SL1. Confirming our expectation, *TxFee* registers a positive effect in Model SL0 (1.777) – a higher fee makes confirmation in the next block more likely. This positive effect (0.859) reduces considerably after the inclusion of *VisPoolRank* and *HigherFeeArrival* in Model SL1, implying that the two priority proxy variables explain a large portion of *TxFee*'s impact on *ConfBlock*. The strong explanatory power of *VisPoolRank* and *HigherFeeArrival* contrast sharply with the ineffectiveness of the simple product term between *TxFee* and *VisPoolNumTx* in Model OLS2 (preliminary analysis), attesting to the value of our theoretical framework. *TxFee* continues to exert a positive impact, while *VisPoolRank* and *HigherFeeArrival* exert a negative impact in Models SL2 and SL3, as our framework posits. For Model LN, where the logged *ConfBlock* is the dependent variable, the respective effects are expected to be in the opposite direction, which is confirmed by our estimation results.

To further strengthen the value of our framework, especially on the use of *VisPoolRank* and *HigherFeeArrival* as proxies for the true rank (see Figure 3), we run an additional regression with *NextBlockRank* as the dependent variable²⁰, which calculates the true rank of the focal transaction in the memory pool at the time when the next block (B_1 in Figure 2) is assembled. Since *VisPoolRank* is a precise measure of the focal transaction's rank in the visible pool, a 1-unit increase is expected to trigger 1-unit increase in *NextBlockRank*. In contrast, *HigherFeeArrival* is expected to have a smaller impact as it proxies but does not precisely measure the focal transaction's invisible pool rank. Model RNB shows highly confirmatory results: *VisPoolRank* and *HigherFeeArrival* register effects of 1.003 and 0.654, respectively.

²⁰ We thank an anonymous reviewer for this suggestion.

Together with a 98.6% R^2 , this provides strong evidence for the structural correctness of our theoretical framework.

Appendix C contains four sets of robustness checks: 1) an alternative measure for *VisPoolRank*, 2) dropping correlated congestion signal controls, 3) a replication sample from the same time period, and 4) another replication sample from a different time period (January 2021 and April 2021). To summarize, we obtain qualitatively consistent results from the first and the third check, while estimates from the second check is almost numerically identical to those in Table 5.

7. Confirmation Time Distribution

Recall from Figure 3 that confirmation time is jointly determined by block of confirmation (*ConfBlock*) and Inter-Block Time (*IBT*). Appendix D discusses why and how *IBT* can be nearly perfectly approximated by an exponential distribution, as assumed in analytical models of Bitcoin systems, with the rate parameter equal to the average *IBT* obtained from data. Combining estimation results from the previous section with *IBT* approximation, Section 7.1 proposes a method to efficiently compute the confirmation time distribution. Applying this method to data, Section 7.2 formally tests hypotheses and draws theoretical contributions. Section 7.3 discusses a practical application of our computational method: recommending transaction fees.

7.1. Computational Procedure

For generality, consider the task of computing the distribution of *ConfTime* under a vector of m *TxFee* values. Our procedure to complete this task can be described in three steps, with each boiling down to the construction of a matrix, as depicted in Figure 6.

$$\begin{array}{ccc}
 \begin{array}{c} \text{K blocks} \\ \left[\begin{array}{ccc} a_{11} & \cdots & a_{1K} \\ \vdots & \ddots & \vdots \\ a_{T1} & \cdots & a_{TK} \end{array} \right] \end{array} & \begin{array}{c} \text{M fee offers} \\ \left[\begin{array}{ccc} b_{11} & \cdots & b_{1M} \\ \vdots & \ddots & \vdots \\ b_{K1} & \cdots & a_{KM} \end{array} \right] \end{array} & \begin{array}{c} \text{M fee offers} \\ \left[\begin{array}{ccc} c_{11} & \cdots & c_{1M} \\ \vdots & \ddots & \vdots \\ c_{T1} & \cdots & c_{TM} \end{array} \right] \end{array} \\
 \left. \begin{array}{c} \text{T seconds} \end{array} \right\} & \left. \begin{array}{c} \text{K blocks} \end{array} \right\} & \left. \begin{array}{c} \text{T seconds} \end{array} \right\} \\
 \alpha_{t,k}: \text{Probability of block } k \text{ arriving at second } t & b_{k,m}: \text{Probability of confirming in block } k \text{ given } m\text{th fee offer} & c_{t,m}: \text{Probability of confirming at second } t \text{ given } m\text{th fee offer}
 \end{array}$$

Figure 6. Matrices for Computing the Confirmation Time Distribution

Step 1: Construct the time-block probability matrix **A**. Each column (row) in **A** represents a specific future block (second) where the focal transaction might be included²¹. The number of blocks to include, K , depends on how far on the tail of the *ConfTime* distribution one wants to estimate. We use $K = 600$ in the examples below since $ConfBlock \leq 600$ covers 99.2% of observations in the data, which should ensure enough tail accuracy for most applications. Correspondingly, we use $T = 600 \times 10 \times 60 = 360000$, where the middle factor (i.e., 10 minutes) is the expected *IBT* rounded to the nearest integer. Again, T could be extended further to improve tail accuracy. Therefore, element $a_{t,k}$ represents the probability of block k arriving at second t . Since the convolution of k exponential distributions follows a Gamma distribution with the shape and rate parameters equal to k and $\frac{1}{E[IBT]}$, the arrival time of block k is distributed as $\Gamma(k, \frac{1}{E[IBT]})$. Given the memoryless property of exponential distribution, the specific time the focal transaction enters relative to previous block (i.e., *TimeSinceB₀*) does not matter for the construction of **A**.

Step 2: Construct the block-fee probability matrix **B**. For each of m fee offers, first compute the corresponding *VisPoolRank* and *HigherFeeArrival* values. Plugging these three variables, together with other control variables, into the estimated *ConfBlock* regressions, we obtain the probability of confirmation in each of the K blocks under a specific fee. These probabilities populate the matrix **B**. Note that the first three rows (i.e., confirmation probability in block 1-3) are computed through Model SL1-3 and the rest through Model LN. For example, the first column of matrix **B** contains the probability of being confirmed in block 1, 2, 3, ..., 600 given the first fee offer.

Step 3: Compute the time-fee probability matrix **C**. Multiplying matrices **A** and **B** yields a new matrix **C**. Element $c_{t,m}$ corresponds to the probability of being confirmed at second t given the m th fee offer. To see this, consider $c_{t,m} = \sum_{k=1}^K a_{t,k} b_{k,m}$. Given the interpretation of $a_{t,k}$ and $b_{k,m}$ (see Figure 6), their product stands for the probability of being confirmed at second t via block k . Since other blocks could also arrive at second t , we need to stack up the block-specific probabilities to obtain the aggregate probability

²¹ Note that to ensure the smoothness of the estimated *ConfTime* distribution, we use seconds on the row-dimension. It could be replaced by minutes if granularity is of less concern.

of confirmation at second t . Then, it follows that column m is essentially the estimated distribution of *ConfTime* under the m th fee offer. That is, each row in column m contains the probability of being confirmed in a specific second under this fee offer.

Next, we implement the above procedure with a small-scale numerical example. To replicate reality as much as possible, we construct three congestion levels (low, medium, and high) by considering a hypothetical transaction joining the memory pool when its size is roughly equal to the three quartiles of *VisPoolNumTx* in our sample (i.e., the transaction-level congestion measure). For *TxFee*, we consider 10, 100, 200, ..., 800 satoshi-per-byte. This reflects a wide range of fees that could potentially be offered by customers of highly heterogenous willingness to wait. To summarize, the three-step procedure is executed three times, once for each congestion scenario, with ten different fee offers (i.e., $M = 9$). Results are shown in Table 6.

Table 6: Estimated Confirmation Time Distribution under Various Congestion Levels (Minutes)

		Input: <i>TxFee</i>								
		10	100	200	300	400	500	600	700	800
		<i>Low Congestion</i>								
Intermediary:	<i>VisPoolRank</i>	836	320	107	74	49	38	34	9	8
	<i>HigherFeeArrival</i>	789	482	73	46	35	21	18	6	3
Output:	<i>ConfBlock Prob.</i>									
	1st Block	61.9%	79.0%	91.2%	94.6%	96.6%	97.9%	98.7%	99.2%	99.5%
	2nd Block	20.0%	14.0%	7.0%	4.6%	3.0%	1.9%	1.2%	0.7%	0.5%
	3rd Block	6.9%	3.3%	1.0%	0.5%	0.2%	0.1%	0.1%	0.0%	0.0%
	>= 4th Block	11.2%	3.8%	0.8%	0.3%	0.1%	0.1%	0.0%	0.0%	0.0%
	<i>ConfTime Dist.</i> (minutes)									
	Mean	30.7	16.0	10.8	10.0	9.6	9.4	9.3	9.3	9.2
	90th percentile	45.2	29.3	23.8	22.6	22.0	21.7	21.5	21.3	21.3
		<i>Medium Congestion</i>								
Intermediary:	<i>VisPoolRank</i>	23625	2659	385	188	124	96	89	9	4
	<i>HigherFeeArrival</i>	2589	1884	247	108	34	14	10	3	1
Output:	<i>ConfBlock Prob.</i>									
	1st Block	4.4%	27.1%	83.7%	90.9%	94.7%	96.8%	98.0%	98.8%	99.2%
	2nd Block	1.7%	20.5%	11.2%	7.1%	4.4%	2.9%	1.8%	1.1%	0.7%
	3rd Block	0.6%	11.6%	2.3%	1.0%	0.5%	0.2%	0.1%	0.1%	0.0%
	>= 4th Block	93.3%	40.8%	2.8%	1.0%	0.4%	0.1%	0.1%	0.0%	0.0%
	<i>ConfTime Dist.</i> (minutes)									
	Mean	647.1	104.0	13.3	10.8	10.0	9.6	9.4	9.3	9.3
	90th percentile	1767.6	247.9	27.1	23.9	22.6	22.0	21.7	21.5	21.3
		<i>High Congestion</i>								
Intermediary:	<i>VisPoolRank</i>	56252	30242	3962	1787	1286	325	120	58	46
	<i>HigherFeeArrival</i>	2263	2241	2041	1821	1610	1067	103	42	28
Output:	<i>ConfBlock Prob.</i>									
	1st Block	0.7%	2.7%	13.3%	26.2%	43.4%	71.9%	93.4%	96.1%	97.6%
	2nd Block	0.0%	0.8%	15.9%	23.7%	25.6%	18.4%	5.6%	3.4%	2.2%
	3rd Block	0.0%	0.1%	7.2%	8.6%	7.0%	3.2%	0.5%	0.2%	0.1%
	>= 4th Block	99.3%	96.4%	63.6%	41.5%	24.0%	6.5%	0.6%	0.2%	0.1%
	<i>ConfTime Dist.</i> (minutes)									
	Mean	2013.7	1077.5	272.6	145.1	76.0	23.5	10.4	9.8	9.5
	90th percentile	6000.0	3474.6	717.2	367.2	157.1	34.3	23.0	22.2	21.7

7.2. Hypothesis Testing

Note that we label *VisPoolRank* and *HigherFeeArrival* as “intermediary” since they are deterministic functions of a fee offer and the current congestion characteristics. The *ConfBlock* probability results are produced from Step 2 of the computation procedure, while *ConfTime* results are generated from Step 3. This presentation flow aligns with our theoretical framework in Figure 3. We first gather evidence for congestion dependence. Consider changes in the mean *ConfTime* (second last row in each panel). As *VisPoolRank* and *HigherFeeArrival* decrease (triggered by the increase in *TxFee* from 10 to 800), we observe a consistent acceleration of *ConfTime*, lending support for H1a and H1b. We also observe that such acceleration results from the ability to improve a transaction’s probability to be included in earlier blocks (visible by how the pattern of *ConfBlock* probability shifts). Further, the acceleration is disproportionately more pronounced in high-congestion and low-fee scenarios. For example, under high congestion, a 10-to-100 satoshi-per-byte increase in *TxFee* results in a mean *ConfTime* decrease from 2013.7 to 1077.5 minutes. In contrast, once *TxFee* is above a certain threshold, further increase does not lower *ConfTime* anymore since the transaction is most likely to be included in the 1st future block. This threshold is about 200/400/600 satoshi-per-byte under the low/medium/high congestion scenarios, respectively.

Next, we gather evidence for the tail shrinkage effect by comparing the changes in the 90th percentile of *ConfTime* (last row of each panel) with changes in its mean. Note that the 3-step computation, once carried out, allows tabulation of any percentile. The 90th percentile is chosen for illustration purposes. When *TxFee* is below the threshold mentioned above and makes a nontrivial impact on *ConfTime*, the 90th percentile always decreases in a larger magnitude than the mean. As a result, while a higher *TxFee* shortens mean *ConfTime*, it also shrinks the tail of the distribution, lending support for H2. Another interesting observation is that the 90th percentile does not go below 20 minutes even with a very high *TxFee* (e.g., 800 satoshi-per-byte), regardless of the congestion level. This is because even if a high enough *TxFee* ensures inclusion in the 1st future block, randomness in *ConfTime* still comes from IBT as our theoretical framework indicates.

To summarize, both H1 and H2 are supported. In addition, the above insights regarding the bitcoin fee-speed relationship are much richer than what the black-box style preliminary analysis in Section 5 can offer.

Attaining these insights, in turn, relies on the development of a theoretical framework firmly grounded in the Bitcoin context.

7.3. Fee Recommendation

The three-step computational procedure is efficient enough to be used as a fee recommendation tool. First, it is very fast – enough to support “real-time” implementation. Note that Step 1 is not transaction-specific and could be completed before a transaction arrives. The main computational complexity for Step 2 comes from the calculation of *VisPoolRank* and *HigherFeeArrival* (as well as other variables that require looping), while Step 3 merely involves the multiplication of two matrices. With an Intel Core i5-8400 processor, steps 2 and 3 together take approximately 3 seconds for $M = 30$. With commercial computing power, we believe an even larger m could be computed with negligible delay.

Second, the format of the final output, matrix **C**, makes the construction of a fee recommendation system straightforward. As discussed in Section 3.2, customers in congested service settings are more interested in the target service level than the mean waiting time. This translates into a Bitcoin user inquiry such as “*if I want my transaction to be confirmed in 30 minutes with a 90% likelihood, how much fee should I offer?*” With a large enough M (i.e., including a wide range of *TxFee* values), the answer to this question can be obtained with a simple search after first converting the probabilities in **C** into cumulative probabilities (i.e., the probability of being confirmed at or before second t) – for instance, by locating row 1800 (30 minutes = 1800 seconds) and then finding the column that contains a number closest to 0.9. If it is found at the m th column, the m th fee offer should be suggested.

8. Conclusion

Cryptocurrencies such as Bitcoin are revolutionary technologies that offer a decentralized and anonymous way to transfer payments. Compared with traditional money transfer services, Bitcoin transactions, on average, have considerably lower fees than wire transfers and much faster speeds than ACH transfers. Furthermore, unlike wire and ACH where a fee-speed schedule is predetermined by the financial institution, Bitcoin users can name their own prices or even choose to not pay any fee at all. Such benefits,

however, come with certain trade-offs such as high volatility of transaction speed and the lack of transparency in the fee-speed relationship, which are consequences of the inherent Bitcoin protocol design. In this study, we adopt a process analysis perspective to unpack this nuanced fee-speed relationship among Bitcoin transactions.

8.1. Theoretical Contributions

In his editorial piece on the scope of OM discipline, Browning (2020, p.496) argues that “*Working processes need to be discovered, described, and understood before they can be improved, controlled, and prescribed. Quite a bit of work is needed merely to **describe** some of the important activities, practices, processes, and operating systems utilized in diverse organizations. Only then can we begin to sink our teeth into developing better theories about how best to manage them.*”

The blockchain technology – in particular its cryptocurrency application – is widely regarded as an Industry 4.0 era disruptive technology that will have tremendous implications on productions and operations management (Choi et al., 2022). Yet, despite its first introduction by Nakamoto (2008) over a decade ago and its growing popularity since then, questions such as how blockchain transaction confirmation process works have been scantily investigated. Our paper aims to fill this gap by providing concise descriptions on various aspects of the Bitcoin system (the cryptocurrency of interest in this paper), such as the block size limit and miner’s incentive structure. We further develop a framework that maps the causal path of transaction fee’s impact on transaction confirmation times. The proposed framework depicts Bitcoin as a priority queueing system with flexible fees. Accordingly, it both draws on and contributes to two longstanding OM literature streams: NYOP and congested service pricing. While prior literature suggests that the magnitude of the fee-speed relationship likely depends on the system congestion level, the exact form of and the mechanisms behind this dependence would only come to light through our framework. We compare two preliminary analyses, which demonstrate what one would do without the proposed framework, with a set of framework-based analyses. The latter produces much richer results, especially regarding how congestion dependence is captured. The simplicity and clarity of our framework should facilitate future operational analyses of cryptocurrency systems.

Babich and Hilary (2020) argue that classic service OM concepts can be instrumental in analyzing Bitcoin confirmation times. Our second theoretical contribution comes from the application of one of these concepts – i.e., delay announcements. As demonstrated in congested service settings such as healthcare (Dong et al., 2019) and call centers (Akşın et al., 2017; Yu et al., 2016), announcement of waiting delays has positive effects on service quality outcomes. Akin to the mechanisms that forecast delays in those settings, the computational procedure proposed in this paper allows for an estimation of the waiting (i.e., confirmation) time distribution for a given range of fees. Accordingly, it expands the prior literature on delay announcement mechanisms to a new context – i.e., the Bitcoin network.

8.2. Practical Implications

From a practical standpoint, our study should be of great interest to various Bitcoin stakeholders. The first beneficiary is regular users – i.e., transaction senders. It is evident that the lack of transparency in the fee-speed relationship can lead to overpayments in transaction fees (Easley et al., 2019). The scale of these overpayments is not trivial; we estimate that a Bitcoin user, on average, pays 48% more than the minimum fee needed to receive the same service (i.e., enter the same block)²². While it is impossible to perfectly predict the future state of the network, hence the exact confirmation block, the three-step procedure proposed in this study is a stepping-stone for improving the match between senders' speed preference and the potential fees needed for confirmation. Specifically, it takes a range of transaction fees as input and computes the resulting set of confirmation time distributions. This approach has obvious advantages over the existing average-time-based fee setting alternatives (e.g., Bitcoin Core Wallet) as users can name a target service level and be recommended a fee. Considering the large and rapidly growing total amount of fees paid (over \$326 million in 2020), we believe improvements in fee recommendation have substantial customer welfare ramifications.

²² Based on a calculation from 1 million randomly sampled transactions. For each transaction, we estimate overpayment as the difference between the actual fee paid and the lowest fee paid within the same block (after trimming the bottom 10% fees).

Next, enhancing transparency into the fee-speed relationship also has long-run implications for the sustainability of the Bitcoin transaction system. Recall from Section 2.3 that the Bitcoin protocol dictates minting rewards to be halved roughly every four years, which gradually shifts the reliance of miner incentive scheme to the transaction fee revenue. Thus, it is to miners' benefit to gain more insights into how much extra speed an increase in fee is able to produce.

Last, a longstanding discussion in the Bitcoin developer community is whether the block size limit should be relaxed to improve system processing capacity. Despite its evident capacity and hence consumer welfare benefits, a larger block size also fundamentally changes the current system congestion dynamics as it disincentivizes users to pay transaction fees. This concern has been grave enough to have instigated heated debates between influential software developers in the cryptocurrency ecosystem and eventually led to the creation of new currency systems such as Bitcoin Cash (Bogart, 2017). Our fee-speed findings are important inputs for the scalability debate (Price and Williams-Grut, 2017) and should be useful for the future development of cryptocurrency functionality. Overall, we believe more operational transparency on the fee-speed relationship creates a win-win-win situation for Bitcoin users, miners, and developers – a goal that many blockchain applications strive to achieve (Choi and Luo, 2019).

8.3. Limitations and Future Work

We caution that our results must be interpreted within the boundaries of this research. One limitation is the use of data from a single cryptocurrency system – i.e., Bitcoin. While it is plausible for our findings to be applicable to other cryptocurrencies that follow the same transaction fee mechanism as the Bitcoin system (e.g., Bitcoin Cash), future research can consider the heterogeneity across systems and investigate if there exist any unique features that can alter the fee-speed relationship in different settings. A second limitation is that per protocol, the Bitcoin transaction data does not include an authentic timestamp that identifies the precise transmission time of the transaction. However, this time can be proxied using the timestamp when the transaction has been received by a major mining node in the Bitcoin network. Calculating the difference between this timestamp and the blockchain publication timestamp of the block that the transaction was included in is a common approach to estimate the confirmation time of a transaction

(Easley et al., 2019; Möser and Böhme, 2015), which we also follow. Future research can consider exploring different approaches to pinpoint transmission timestamps.

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Appendix A: Correlation Matrix

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)
(a) ConfTime																				
(b) ConfBlock	1.0																			
(c) TxFee	-0.08	-0.08																		
(d) VisPoolRank	0.42	0.41	-0.1																	
(e) VisPoolUptoSize	0.54	0.54	-0.11	0.8																
(f) HigherFeeArrival	0.22	0.22	-0.32	0.41	0.32															
(g) VisPoolNumTx	0.06	0.05	0.54	0.27	0.16	0.02														
(h) VisPoolAvgFee	0.06	0.06	0.61	0.23	0.14	0.01	0.81													
(i) PastAvgFeeBlock1	0.05	0.05	0.69	0.17	0.12	0.04	0.77	0.85												
(j) PastAvgFeeBlock2	0.06	0.05	0.66	0.19	0.12	0.03	0.81	0.89	0.94											
(k) PastAvgFeeBlock3	0.05	0.05	0.6	0.2	0.12	0.02	0.8	0.88	0.86	0.92										
(l) PastAvgFeeBlock4	0.06	0.05	0.58	0.2	0.12	0	0.79	0.87	0.83	0.89	0.91									
(m) PastAvgFeeBlock5	0.05	0.05	0.56	0.2	0.12	0.01	0.78	0.85	0.79	0.85	0.86	0.88								
(n) PastAvgFeeBlock6	0.06	0.05	0.57	0.2	0.12	0	0.8	0.87	0.8	0.88	0.88	0.9	0.89							
(o) TimeSinceB0	0.02	0.02	-0.01	0.12	0.07	0	0.06	0.17	0.01	0	0.01	0	0	-0.01						
(p) TxSize	0.04	0.05	-0.04	0.03	0.07	0.04	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	0					
(q) ExchangeRate	0.05	0.06	0.32	0.06	0.05	0.12	0.39	0.48	0.41	0.47	0.46	0.42	0.44	0.45	-0.05	-0.02				
(r) NextBlockRank	0.43	0.41	-0.11	0.99	0.79	0.44	0.28	0.23	0.18	0.2	0.21	0.21	0.2	0.21	0.11	0.03	0.07			
(s) TxValue	-0.01	-0.01	0.01	-0.01	-0.01	-0.01	0	0	0	0	0	0	0	0	0.01	0.03	-0.01	-0.01		
(t) TxValueVPR	0.1	0.09	0.29	0.27	0.17	0.06	0.66	0.5	0.46	0.49	0.49	0.49	0.48	0.5	0.02	-0.05	0.2	0.27	-0.04	
(u) TxValueHFA	0.1	0.1	0.04	0.13	0.09	0.22	0.21	0.18	0.14	0.18	0.19	0.19	0.18	0.2	-0.08	-0.09	0.23	0.14	-0.09	0.63

Appendix B: Preliminary Analysis Full Regression Results

	Model OLS1	Model OLS2	Model QR1	Model QR2	Model QR3	Model QR4	Model QR5	Model QR6	Model QR7	Model QR8	Model QR9
Intercept	2.266*** (0.0102)	2.264*** (0.0102)	0.702*** (0.0122)	1.161*** (0.0113)	1.471*** (0.0103)	1.776*** (0.0098)	2.122*** (0.0101)	2.449*** (0.0100)	2.792*** (0.0107)	3.309*** (0.0126)	4.359*** (0.0198)
<i>TxFee</i> $\times 10^{-2}$	-0.176*** (0.0005)	-0.181*** (0.0007)	-0.068*** (0.0006)	-0.113*** (0.0005)	-0.144*** (0.0005)	-0.176*** (0.0005)	-0.209*** (0.0005)	-0.245*** (0.0005)	-0.287*** (0.0005)	-0.338*** (0.0006)	-0.39*** (0.0009)
<i>VisPoolNumTx</i> $\times 10^{-3}$	0.004*** (0.0001)	0.004*** (0.0001)	0.001*** (0.0001)	0.001*** (0.0001)	0.002*** (0.0001)	0.003*** (0.0001)	0.003*** (0.0001)	0.004*** (0.0001)	0.005*** (0.0001)	0.006*** (0.0001)	0.011*** (0.0001)
<i>TxFee</i> x <i>VisPoolNumTx</i> $\times 10^{-5}$		0.00006*** (0.000007)									
<i>MempoolAvgFee</i> $\times 10^{-2}$	0.318*** (0.0051)	0.316*** (0.0051)	0.06*** (0.0061)	0.08*** (0.0057)	0.111*** (0.0052)	0.175*** (0.0049)	0.218*** (0.0051)	0.26*** (0.0050)	0.292*** (0.0054)	0.378*** (0.0063)	0.544*** (0.0100)
<i>PastAvgFeeBlock1</i> $\times 10^{-2}$	0.047*** (0.0015)	0.048*** (0.0015)	0.051*** (0.0018)	0.079*** (0.0017)	0.093*** (0.0015)	0.107*** (0.0014)	0.127*** (0.0015)	0.15*** (0.0015)	0.165*** (0.0016)	0.175*** (0.0019)	0.118*** (0.0029)
<i>PastAvgFeeBlock2</i> $\times 10^{-2}$	0.006** (0.0021)	0.006** (0.0021)	-0.008** (0.0025)	0.001 (0.0023)	0.003 (0.0021)	-0.001 (0.0020)	-0.009*** (0.0021)	-0.011*** (0.0021)	0.008*** (0.0022)	0.031*** (0.0026)	-0.003 (0.0041)
<i>PastAvgFeeBlock3</i> $\times 10^{-2}$	0.004* (0.0016)	0.003* (0.0016)	-0.001 (0.0019)	0.008*** (0.0017)	0.019*** (0.0016)	0.021*** (0.0015)	0.02*** (0.0015)	0.022*** (0.0015)	0.016*** (0.0016)	-0.002 (0.0019)	0.007* (0.0030)
<i>PastAvgFeeBlock4</i> $\times 10^{-2}$	0.013*** (0.0015)	0.013*** (0.0015)	-0.02*** (0.0018)	-0.021*** (0.0017)	-0.021*** (0.0016)	-0.015*** (0.0015)	-0.011*** (0.0015)	-0.016*** (0.0015)	-0.035*** (0.0016)	-0.039*** (0.0019)	0.027*** (0.0030)
<i>PastAvgFeeBlock5</i> $\times 10^{-2}$	0.033*** (0.0014)	0.033*** (0.0014)	0.046*** (0.0016)	0.042*** (0.0015)	0.038*** (0.0014)	0.027*** (0.0013)	0.019*** (0.0014)	0.015*** (0.0013)	0.028*** (0.0014)	0.049*** (0.0017)	0.057*** (0.0027)
<i>PastAvgFeeBlock6</i> $\times 10^{-2}$	-0.033*** (0.0018)	-0.034*** (0.0018)	-0.021*** (0.0021)	-0.023*** (0.0020)	-0.028*** (0.0018)	-0.031*** (0.0017)	-0.022*** (0.0018)	-0.016*** (0.0017)	-0.012*** (0.0019)	-0.014*** (0.0022)	-0.051*** (0.0035)
<i>TimeSinceB₀</i>	0.017*** (0.0001)	0.017*** (0.0001)	0.007*** (0.0002)	0.012*** (0.0002)	0.015*** (0.0001)	0.017*** (0.0001)	0.018*** (0.0001)	0.019*** (0.0001)	0.02*** (0.0001)	0.02*** (0.0002)	0.019*** (0.0003)
<i>TxSize</i>	0.032*** (0.0004)	0.032*** (0.0004)	0.006*** (0.0004)	0.01*** (0.0004)	0.014*** (0.0004)	0.018*** (0.0004)	0.024*** (0.0004)	0.031*** (0.0004)	0.04*** (0.0004)	0.053*** (0.0005)	0.08*** (0.0007)
<i>ExchangeRate</i>	-0.023*** (0.0006)	-0.021*** (0.0006)	-0.003*** (0.0007)	-0.004*** (0.0007)	-0.004*** (0.0006)	-0.009*** (0.0006)	-0.017*** (0.0006)	-0.025*** (0.0006)	-0.03*** (0.0006)	-0.042*** (0.0007)	-0.071*** (0.0012)
Fixed Effects											
<i>Month</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Day of Week</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Hour</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

*p<0.05, **p<0.01, ***p<0.001

$\times 10^{-n}$ means the coefficient is scaled up by a factor of 10^n for exposition purpose.

Month/Day of Week/Hour contains 4/6/23 dummy variables to capture the heterogeneity across months, days of the week, and hours of the day.

Appendix C: Robustness Check Results

We carry out four sets of robustness checks, with each re-running Model SL1-3 and LN. The first set considers an alternative measure for *VisPoolRank*: the cumulative file size of transactions in the t_l memory pool with *TxFee* higher than the focal transaction, *VisPoolUptoSize*. We prefer *VisPoolRank* in the main analysis for its straightforward interpretation. The advantage of *VisPoolUptoSize*, however, is that it is closer to a profit-maximizing miner's transaction selection rule: making a cut when *VisPoolUptoSize* reaches the block capacity. Results are reported in the Panel A of the table below. Coefficients on *TxFee* and *HigherFeeArrival* are similar to those in Table 5. While the effect of *VisPoolUptoSize* cannot be directly compared with that of *VisPoolRank*, they are qualitatively consistent such that a larger size and a higher rank both reduce the chance of being included in the next block.

The second robustness test keeps only *VisPoolNumTx* as the congestion signal control. Recall from the *Congestion Signal Controls* discussion in Section 6.2 that we have used four congestion signal measures to proxy what Bitcoin users might consider when setting fees. As shown in the correlation matrix in Appendix A, the first three measures – *VisPoolNumTx*, *VisPoolAvgFee*, and *PastAvgFeeBlockN* variables – have high correlations. This is expected as all three metrics respond to the level of system congestion. Panel B reports the regression results after dropping *VisPoolAvgFee* and the six *PastAvgFeeBlockN* variables from the list of controls. Its high consistency with Table 5 alleviates any concern that the high correlations among these variables might cause.

The third robustness test considers a replication sample from the same 4-month time period from November 11, 2017 to March 10, 2018. Recall from Section 5 that we have split our full sample for the feasibility of running regressions. Analyses in the body text were run on the first sample. Panel C shows results from the second sample, where coefficient estimates are almost identical to those in Table 5, giving strong confidence on our sampling randomness.

The fourth robustness test collects a new data set from a more recent 4-month time period from January 2021 to April 2021, following the same data collection approach as outlined in Section 4.1. A total of 1,200 blocks are sampled (i.e., 10 blocks per day times 120 days), yielding 2,381,293 transactions. This is 11% more than the 2,141,297 transactions in the main analysis, affirming the general observations that the congestion level in the Bitcoin system is increasing. A comparison of the Panel D estimates with those in Table 5 of the manuscript reveals qualitatively consistent results – i.e., the direction and the statistical significance of the coefficients are preserved. The effect sizes are, however, larger than those in Table 5, which is aligned with the expectation that the impact of transaction fee is more salient when the system is more congested.

Table: Robustness Check Results

DV	Model SL1	Model SL2	Model SL3	Model LN
Sample	I($CB = 1$)	I($CB = 2$)	I($CB = 3$)	log(CB)
	All Obs.	$CB > 1$	$CB > 2$	$CB > 3$
Panel A: Alternative Measure for <i>VisPoolRank</i>				
<i>TxFee</i>	0.852***	0.738***	0.541***	-0.37***
$\times 10^{-2}$	(0.014)	(0.024)	(0.035)	(0.022)
<i>VisPoolUptoSize</i>	-0.034***	-0.274***	-0.279***	0.093***
	(0.004)	(0.004)	(0.005)	(0.002)
<i>HigherFeeArrival</i>	-2.114***	-0.641***	-0.19***	0.973***
$\times 10^{-3}$	(0.031)	(0.038)	(0.057)	(0.031)
N	2,141,297	669,496	421,552	327,580
LL	-905675	-357941	-188532	
R ²				43.10%
Panel B: Only <i>VisPoolNumTx</i> as Congestion Signal Control				
<i>TxFee</i>	0.85***	0.751***	0.469***	-0.287***
$\times 10^{-2}$	(0.015)	(0.024)	(0.034)	(0.023)
<i>VisPoolRank</i>	-0.016***	-0.155***	-0.215***	0.053***
$\times 10^{-3}$	(0.003)	(0.003)	(0.004)	(0.002)
<i>HigherFeeArrival</i>	-2.161***	-0.641***	-0.136*	1.298***
$\times 10^{-3}$	(0.031)	(0.038)	(0.057)	(0.032)
N	2,141,297	669,131	420,418	327,156
LL	-907046	-361066	-187619	
R ²				37.50%
Panel C: Replication Sample (Same Time Period as Table 4)				
<i>TxFee</i>	0.85***	0.751***	0.469***	-0.287***
$\times 10^{-2}$	(0.015)	(0.024)	(0.034)	(0.023)
<i>VisPoolRank</i>	-0.016***	-0.155***	-0.215***	0.053***
$\times 10^{-3}$	(0.003)	(0.003)	(0.004)	(0.002)
<i>HigherFeeArrival</i>	-2.161***	-0.641***	-0.136*	1.298***
$\times 10^{-3}$	(0.031)	(0.038)	(0.057)	(0.032)
N	2,141,297	669,131	420,418	327,156
LL	-907046	-361066	-187619	
R ²				37.50%
Panel D: New Sample January 2021 to April 2021				
<i>TxFee</i>	2.707***	3.138***	3.308***	-1.482***
$\times 10^{-2}$	(0.027)	(0.058)	(0.132)	(0.097)
<i>VisPoolRank</i>	-0.115***	-0.128***	-0.119***	0.068***
$\times 10^{-3}$	(0.003)	(0.004)	(0.004)	(0.002)
<i>HigherFeeArrival</i>	-2.457***	-2.164***	-2.219***	2.327***
$\times 10^{-3}$	(0.022)	(0.044)	(0.090)	(0.050)
N	2381293	1005836	675915	534,462
LL	-1190690	-533796	-306201	
R ²				37.20%

A full set of control variables (same as those shown in Table 5) is included in all regressions for estimation yet omitted from the table for brevity. All other Table 5 footnotes apply here.

Appendix D: Approximating Inter-Block Time by Exponential Distribution

Analytical models of the Bitcoin system often assume for simplicity that *IBT* is exponentially distributed. In the following, we provide a context-based reasoning for why this is a realistic assumption and then show empirical evidence confirming this expectation. First, as explained in Section 2.2, most of *IBT* is made up by the hashing step of mining. Second, speed of hashing depends on the total computational power of the entire miner population (relative to the current difficulty level), regardless of the actual number of miners. That is, 10 miners with 1 unit of computing power each is the same as 1 miner with 10 units of power. Third, for the “aggregate miner”, hashing is a trial-and-error process, which ends when one success is reached. If each trial has λ probability of success, probability of solving the hashing problem at the n^{th} trial is $(1 - \lambda)^{n-1}\lambda$. If each trial consumes t seconds, density function of hashing time is essentially $t(1 - \lambda)^{n-1}\lambda$, which is a discrete-time analogue of the continuous exponential distribution. In left panel of the following figure, we plot the histogram of sample *IBT* (using all 18,953 blocks) against the fitted exponential distribution (rate = 0.109). In the right panel, the empirical CDF is plotted against the fitted distribution’s CDF, which indicates a near perfect fit.

